

An arms race between 5'ppp-RNA virus and its alternative recognition receptor MDA5 in RIG-I-lost teleost fish

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Shang Geng, Xing Lv, Weiwei Zheng, Tianjun Xu 

Laboratory of Fish Molecular Immunology, College of Fisheries and Life Science, Shanghai Ocean University, Shanghai, China • Laboratory for Marine Biology and Biotechnology, Qingdao Marine Science and Technology Center, Qingdao, China • Marine Biomedical Science and Technology Innovation Platform of Lin-gang Special Area, Shanghai, China

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Abstract

The incessant arms race between viruses and hosts has led to numerous evolutionary innovations that shape life's evolution. During this process, the interactions between viral receptors and viruses have garnered significant interest since viral receptors are cell surface proteins exploited by viruses to initiate infection. Our study sheds light on the arms race between the MDA5 receptor and 5'ppp-RNA virus in a lower vertebrate fish, *M. miiuy*. Firstly, the frequent and independent loss events of RIG-I in vertebrates prompted us to search for alternative immune substitutes, with homology-dependent genetic compensation response (HDGCR) being the main pathway. Our further analysis suggested that MDA5 of *M. miiuy* and *G. gallus*, the homolog of RIG-I, can replace RIG-I in recognizing 5'ppp-RNA virus, which may lead to redundancy of RIG-I and loss from the species genome during evolution. Secondly, as an adversarial strategy, 5'ppp-RNA SCR can utilize the m⁶A methylation mechanism to degrade MDA5 and weaken its antiviral immune ability, thus promoting its own replication and immune evasion. In summary, our study provides a snapshot into the interaction and coevolution between vertebrate and virus, offering valuable perspectives on the ecological and evolutionary factors that contribute to the diversity of the immune system.

eLife assessment

The authors present evidence suggesting that MDA5 can substitute as a sensor for triphosphate RNA in a species that naturally lacks RIG-I. The key findings are potentially **important** for our understanding of the evolution of innate immune responses. Compared to an earlier version of the paper, the strength of evidence has improved but it is still partially **incomplete** due to a few key missing experiments and controls.

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1. Introduction

The Pattern recognition receptors (PRRs) recognize conserved pathogen-associated molecular pattern (PAMP) motifs, including proteins, lipids and nucleotides, resulting in activation of innate immune responses of host¹. Retinoic acid-induced RIG-I-like receptors (RLRs) are a specific group of PRRs belonging to the family of DExD/H-box RNA helicases, and they play a crucial role in the innate antiviral response. There are three members in the RLR family: retinoic acid-inducible gene I (RIG-I), melanoma differentiation-associated gene 5 (MDA5), and laboratory of genetics and physiology 2 (LGP2). RIG-I and MDA5 exhibit similar signaling characteristics and structural homology, including two N-terminal caspase-recruitment domains (CARDs), a DExD/H box RNA helicase domain, and a C-terminal repressor domain (RD)². However, LGP2 lacks CARD domain². The specific roles of RIG-I and MDA5 in response to virus stimulation are not redundant: RIG-I detects short blunt-end RNA with a 5'-triphosphate motif (5'ppp-RNA), while MDA5 specifically recognizes long forms of viral dsRNA³⁻⁵. In contrast, the involvement of LGP2 in cytosolic RNA sensing remains a topic of debate. Some studies have proposed that LGP2 is crucial for the production of type I interferon (IFN-I) in response to several RIG-I- and MDA5-dependent viruses⁶, whereas others have described LGP2 as a negative regulator of RIG-I signaling⁷. However, surprisingly, despite the important immune functions of RLR members, they have been lost from the genome in many species including mammals, birds, and fish⁸⁻¹⁰.

5'ppp-RNA is present in a wide variety of RNA viruses, including Flaviviridae, Paramyxoviridae, Coronaviridae, Orthomyxoviridae, and Rhabdoviridae families⁴. These viruses widely infect all kinds of life and cause various harmfulness and serious diseases, and the infected hosts range from the highest vertebrates to the lowest vertebrate, teleost fish^{11, 12}. 5'ppp-RNA viruses are expected to be recognized and trigger an antiviral immune response by RIG-I. However, RIG-I is lacking in many vertebrates, so who then recognizes 5'ppp-RNA in these species? When a gene is functionally inactive or lost, other genes or pathways may undergo homology-dependent genetic compensation response (HDGCR) to maintain organismal balance. The pressure from viral infections has led to the evolution of the immune system, and the loss of RIG-I may be accompanied by functional substitution by homologous genes in order to compensate for the loss of RIG-I function. *Tupaia belangeri*, a mammal lacking RIG-I, has been reported to possess the MDA5 protein, which has undergone positive selection and exhibits the ability to sense Sendai virus (an RNA virus posed as a RIG-I agonist) for inducing type I IFN⁸. This study provides insights into the compensatory mechanism of RIG-I deficiency in vertebrates, however, considering the dynamic nature of host-virus dynamics, further research is urgently needed to validate and expand these insights.

For millions of years, viruses have coevolved with their hosts, leading to the development of various mechanisms for immune evasion in response to the mutual evolutionary pressure¹³. Recently, it is reported that N6-methyladenosine (m⁶A) modification plays a crucial role in enabling viruses to evade the interferon (IFN) response¹⁴. m⁶A methylation is an important epigenetic modification that involves the dynamic and reversible process of RNA methylation, regulated by methylase (writer), demethylase (eraser), and reader proteins that recognize m⁶A-modified RNA¹⁵. N6-methylation deposition on mRNA primarily relies on the enzymatic “writing” complex within the cell, with a key composition featuring methyltransferase-like 3 (METTL3) and METTL14. Conversely, N6-demethylation is chiefly modulated by “eraser” proteins, with fat mass and obesity-associated protein (FTO) and AlkB homolog 5 (ALKBH5) being the identified enzymes for this role^{16, 17}. The functional impact of N6-methylation on RNA characteristics typically remains latent until recognition by “reader” proteins, such as members of the YTHDF protein family, encompassing YTH domain family 1 (YTHDF1), YTHDF2, and YTHDF3. In mammals, YTHDF2 and YTHDF3 are recognized for their involvement in the degradation of m⁶A-modified RNA targets, as well as their association with mRNA localization and phase

separation ¹⁸. By utilizing these m⁶A-related enzymes in the host, viruses can not only use m⁶A modification to hide themselves and avoid recognition by host receptors but also manipulate host gene expression to enable immune evasion ^{19–21}.

Fish and birds are commonly observed to exhibit “RIG-I deficiency” phenomena, and they are also the most abundant groups of aquatic and terrestrial vertebrates respectively, giving them certain evolutionary advantages. As a result, the research of the immune evolution of them helps us to understand the immune evolution of the entire vertebrates. Meanwhile, fish and birds are major source of high-quality protein for human consumption, viral diseases in aquaculture and livestock industry pose serious obstacles to the industry’s healthy development. Among them, the 5’ppp-RNA virus, as mentioned earlier, is a highly pathogenic group. Within this group, *siniperca cheats* rhabdovirus (SCRV) is an important pathogen that can cause huge economic losses and serious threats to aquaculture in freshwater and marine fish ²². In addition, vesicular stomatitis virus (VSV) is also a highly infectious and widely prevalent 5’ppp-RNA virus ¹². Therefore, in this study, we selected teleost fish miiuy croaker (*Miichthys miiuy*) and bird chicken (*Gallus gallus*), two vertebrates lacking RIG-I, as well as two 5’ppp-RNA viruses (SCRV and VSV), to investigate the evolutionary strategies and interactions between 5’ppp-RNA virus and host. The final results demonstrated that the MDA5 of *M. miiuy* and *G. gallus* lacking RIG-I both evolved the ability to recognize 5’ppp-RNA. In addition, SCR virus can in turn utilize the m⁶A strategy to degrade MDA5 of *M. miiuy* for immune evasion. In summary, these findings shed light on the functional diversity of innate antiviral activity and unveil a complex arms race between virus replication and the innate immunity of its reservoir hosts in vertebrates. In addition, since the loss of RIG-I in vertebrates is a widespread event, our research also provides a new perspective for vertebrate groups that have lost RIG-I.

2. Results

2.1 A genetic compensation response model for RIG loss

In previous studies, it has been reported that a few vertebrates lost the RIG-I gene during evolution ^{9, 10, 23}. To investigate this further, we utilized the recent advancements in available and comprehensive genomic data of vertebrates from the NCBI and Ensemble websites. By analyzing the genomes of various species, we determined the presence and absence of the RIG-I gene. As depicted in **Figure 1A**, the absence of the RIG-I gene does not appear to be uncommon in vertebrates, with at least three independent loss events observed in mammals, birds and fish, respectively. These gene-loss events occurred over a long evolutionary timescale. Based on the minimum age of the common ancestor of RIG-I-missing species (<http://www.timetree.org/>), the oldest recorded losses occurring in species like *Erpetoichthys calabaricus*, can be traced back to approximately 330 million years ago (MYA). Conversely, the detected loss in *Gallus gallus* took place less than 42 MYA. To gain a deeper understanding of the evolutionary dynamics of vertebrate RIG-I, we explored the conservation of the genes surrounding RIG-I. We analyzed the genome sequences of *Homo sapiens*, *Callorhinchus milii*, *Danio rerio*, *Tupaia belangeri*, *Gallus gallus*, and *M. miiuy* (**Figure 1B**). Similar to the slow-evolving *Callorhinchus milii*, humans possess RIG-I with TOPORS and ACO1 as its upstream and downstream genes, respectively. As previously reported ^{8, 10}, *Tupaia belangeri*, *Gallus gallus*, and *M. miiuy* do not have RIG-I. Interestingly, zebrafish has a RIG-I with its upstream gene being TOPORSb, seemingly a result of chromosome fragment duplication. However, one RIG-I gene was lost in the downstream region of TOPORSa, indicating that one of the RIG-I copies has been lost in zebrafish, similar to *M. miiuy*.

Given the important role of RIG-I in recognizing 5’ppp RNA, we selected two different models for experiments to detect the presence of other potential receptors for immune compensation in RIG-I deficient species. Specifically, we extracted virus RNA from SCR and VSV viruses and obtained their dephosphorylated RNA derivatives (SCR-RNA-CIAP and VSV-RNA-CIAP) through Calf

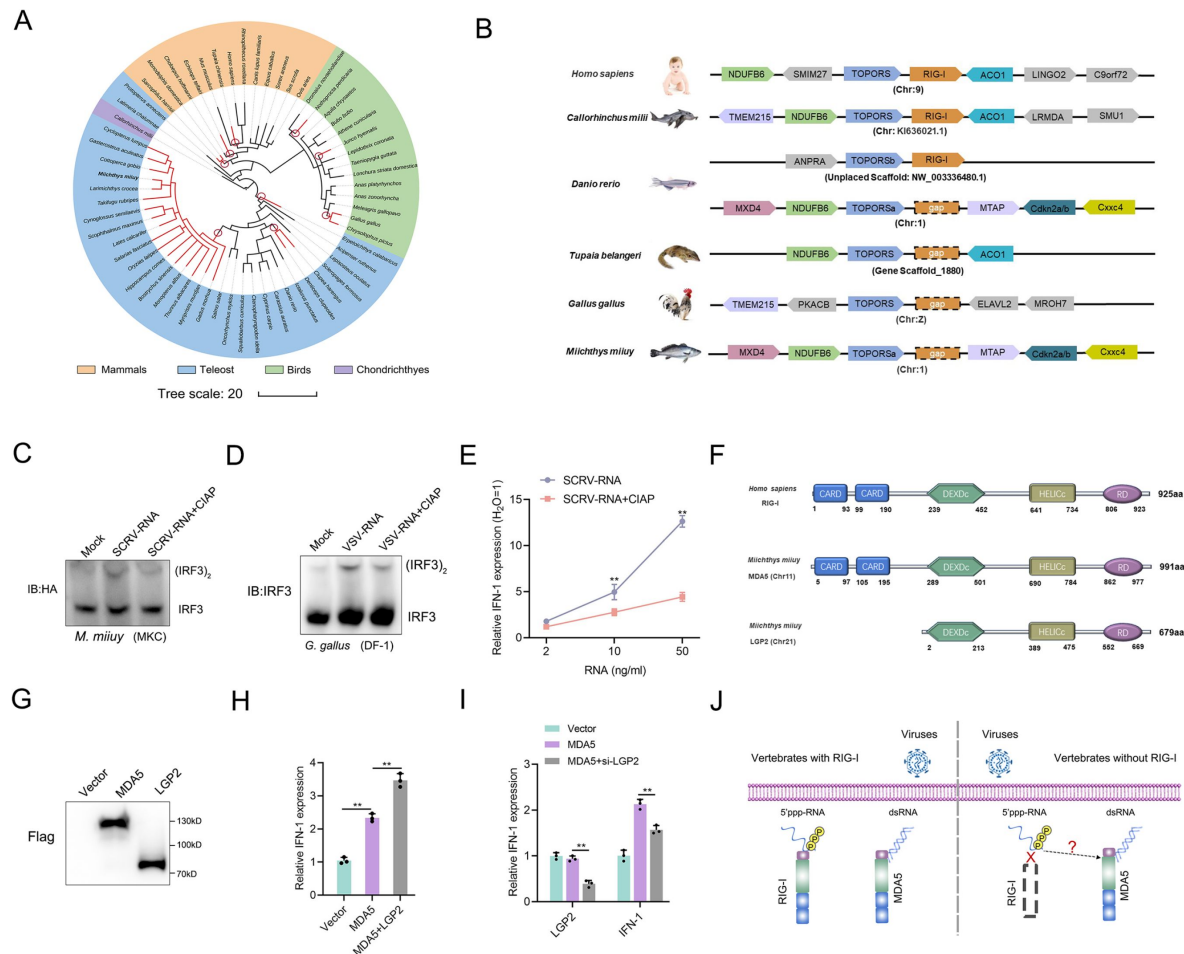


Figure 1.

A genetic compensation response model for RIG loss.

(A) Loss of RIG-I in vertebrates. Each branch tip represents one species. Red lines show lineages where loss of RIG-I. The red circle indicates the occurrence of an independent loss event. Branch lengths scale in millions of years (MYA). (B) Comparative analysis of gene synteny of RIG-I in vertebrate genomes. (C and D) SCR-V-RNA and VSV-RNA were extracted from SCR-V and VSV virus and dephosphorylated them with Calf Intestinal Alkaline Phosphatase (CIAP). Then, MKC cells were transfected with SCR-V-RNA and SCR-V-RNA+CIAP, and DF-1 cells were transfected with VSV-RNA and VSV-RNA+CIAP. IRF3 dimerization was analyzed by native gel electrophoresis. (E) MKC cells were transfected with different concentrations of SCR-V-RNA and SCR-V-RNA+CIAP, then the expression of IFN-1 was detected by qRT-PCR. (F) Predicted protein structures of MDA5 and LGP2 of *M. miiuy* and RIG-I of *H. sapiens*. (G) The expression of MDA5 and LGP2 plasmids of *M. miiuy* were detected by Western Blot. (H and I) MKC cells were co-transfected with MDA5 plasmids and LGP2 plasmids (H) or si-LGP2 (I) for 24 h and then treated with SCR-V (MOI = 5) for 24 h. The expression of IFN-1 was determined by qPCR. (J) A genetic compensation response model for RIG loss. All data presented as the means $\pm$ SE from at least three independent triplicated experiments. **, $p < 0.01$ versus the controls.

Intestinal Alkaline Phosphatase (CIAP) treatment. Next, these RNAs were transfected into MKC cells of *M. miiuy* and DF-1 cells of *G. gallus* respectively to detect the dimerization form of IRF3. Results showed that virus RNA are more potent than dephosphorylated virus RNA in activating IRF3 (**Figure 1C** and **1D**). And further qPCR experiments showed that the induction ability of IFN-1 by SCR-V-RNA is significantly higher than that of SCR-V-RNA-CIAP. These results indicate the presence of viral receptors in *M. miiuy* and *G. gallus* that can recognize 5'ppp-RNA. Next, we compared the structural domains of human RIG-I and *M. miiuy* MDA5 (mmiMDA5) and LGP2 (mmiLGP2) proteins, and observed that mmiLGP2 lacks the CARD domain (**Figure 1F**). We next constructed the mmiLGP2 plasmid and synthesized siRNA targeting mmiLGP2, and co-transfected them with MDA5 plasmids into MKC cells. The results showed that mmiLGP2 can actively regulate mmiMDA5's immune response to the SCR-V virus (**Figure 1H** and **1I**), which further indicates that it participates in antiviral immunity as an intermediate regulatory factor of MDA5 rather than a downstream signal transmitter⁷. Therefore, these results prompted us to investigate whether MDA5, a structurally similar homolog to RIG-I, can substitute for RIG-I in recognizing 5'ppp-RNA in vertebrates without RIG-I (**Figure 1J**).

2.2 MDA5 promotes host antiviral innate immunity

To further investigate the alternative immune function of MDA5, we first investigated the *M. miiuy* MDA5-mediated signaling pathway in response to SCR-V virus and double-stranded mimetic poly(I:C)-HMW infection. Two siRNA molecules (si-MDA5-1 and si-MDA5-2) were designed to assess the function of MDA5. si-MDA5-1 demonstrated higher inhibitory efficiency compared to si-MDA5-2 (**Figure 2A**). Therefore, si-MDA5-1 (referred to as si-MDA5) was selected for subsequent experiments. Subsequently, we constructed the MDA5 expression plasmid and confirmed its protein overexpression efficiency and cytoplasmic localization (**Figure 2B** and **2C**). Considering that mammalian MDA5 activates NF- κ B and IRF3/IRF7 to induce IFNs, we investigated whether *M. miiuy* MDA5 could impact these pathways. Dual-luciferase reporter assays demonstrated that knockdown of MDA5 significantly inhibits NF- κ B, IRF3, IRF7, and IFN-1 reporter genes (**Figure 2D**). In addition, we investigated the role of MDA5 in regulating the expression of IFN-stimulated genes (ISGs), which are crucial effectors. As shown in **Figure 2E-2G**, we observed that knockdown or overexpression of MDA5 can significantly inhibit or promote the expression levels of IFN-1, Mx1, ISG15, and Viperin upon SCR-V infection or poly(I:C)-HMW stimulation. To visually represent the biological significance of MDA5 in SCR-V-induced host cells, we utilized the virus plaque and qPCR methods to examine whether it could influence viral replication. As demonstrated in **Figure 2H-2K**, knockdown of MDA5 led to a significant increase in viral plaque formation and promoted SCR-V replication in SCR-V-infected cells, while overexpression had the opposite effect. Next, we explored that how MDA5 conducts signal transduction, and results showed that MDA5 is capable of binding not only to MAVS but also to STING (**Figure 2L** and **2M**). Under the infection of SCR-V virus, co-transfection of STING plasmids or si-STING with MDA5 plasmids can significantly increase or decrease the induction of IFN-1, indicating the important role of STING in MDA5-mediated immune transmission in response to RNA virus (**Figure 2N** and **2O**). Finally, through the determination of IRF3 dimer and IFN induction, our results showed that the induction of antiviral innate immunity by SCR-V-RNA depends on MDA5 (**Figure 2P** and **2Q**). Overall, these findings indicated that MDA5 plays a positive role in regulating the antiviral responses of *M. miiuy*.

2.3 RD domain is required for MDA5 to recognize SCR-V RNA

We investigated whether *M. miiuy* MDA5, in the absence of RIG-I, can directly bind to the RNA of SCR-V. To address this question, we precipitated the MDA5 protein from SCR-V-infected MKC cells that were transfected with MDA5-Flag and pcDNA3.1-Flag. The bound RNA was then amplified using SCR-V-specific primers or β -actin primers as a control. Our results showed that MDA5 interacted with SCR-V RNA, while β -actin mRNA did not associate with MDA5 (**Figure 3A**). The C-terminal regulatory domain (RD) of human RIG-I binds to viral RNA in a triphosphate-dependent

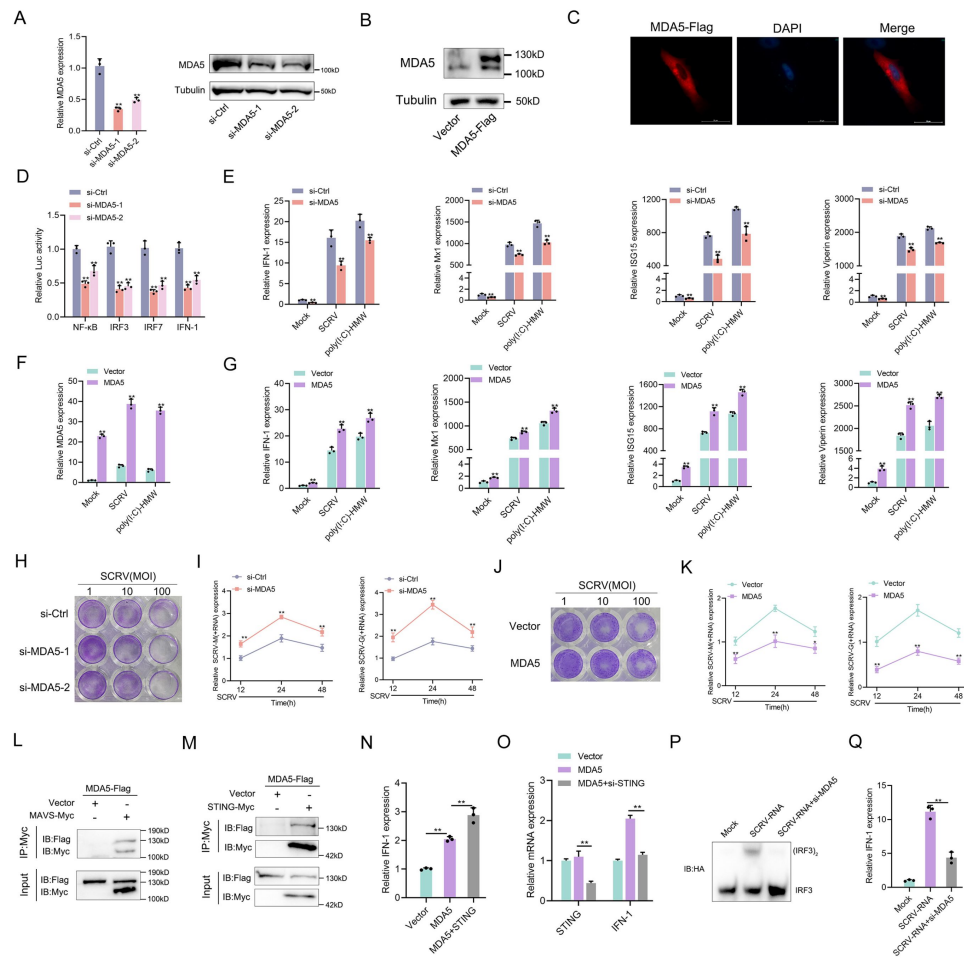


Figure 2.

MDA5 promotes host antiviral innate immunity.

(A) Silence efficiency of si-MDA5 measured by qRT-PCR and Western blotting. Two siRNAs of MDA5 were transfected into MPC cells for 48 h respectively. (B) MKC cells were transfected with MDA5 plasmids for 48 h, then the overexpression efficiency was detected through MDA5 endogenous antibodies. (C) Subcellular localization of MDA5 in MKCs by immunofluorescence. (D) MDA5 knockdown suppresses NF- κ B, IRF3, IRF7, and IFN-1 signaling. MPC cells were transfected with pRL-TK Renilla luciferase plasmid, luciferase reporter genes, together with MDA5 expression plasmid. At 48 h post-transfection, the luciferase activity was measured and normalized to renilla luciferase activity. (E) Knockdown of MDA5 attenuates the expression of antiviral genes. MPC cells were transfected with si-ctrl or si-MDA5 for 24 h and then treated with SCR (MOI = 5) or poly(I:C)-HMW for 24 h. The expression of IFN-1, Mx1, ISG15, and Viperin were determined by qPCR. (F and G) Overexpression of MDA5 promotes the expression of antiviral genes. MKC cells were transfected with vector or MDA5 expression plasmids for 24 h and then treated with SCR (MOI = 5) or poly(I:C)-HMW for 24 h. The expression of MDA5 (F), and antiviral genes, including IFN-1, Mx1, ISG15, and Viperin (G) were determined by qPCR. (H and J) MPC cells transfected with si-ctrl or si-MDA5 and MKC cells transfected with pcDNA3.1 vector or MDA5 plasmid 24 h and then treated with SCR at the dose indicated for 48 h. Then, cell monolayers were fixed with 4% paraformaldehyde and stained with 1% crystal violet. (I and K) MDA5 inhibits SCR replication. MPC cells were transfected with si-ctrl or si-MDA5 and MKC cells were transfected with pcDNA3.1 vector or MDA5 expression plasmid for 24 h, then infected with SCR (MOI = 5) for 12, 24, or 48 h. The qPCR analysis was conducted for SCR-M and SCR-G RNA levels. (L and M) MDA5 immunoprecipitates with MAVS (L) and STING (M). EPC cells ($\sim 1 \times 10^7$) were co-transfected with MAVS-Myc or STING-Myc and MDA5-Flag expression plasmids for 24 h, followed by immunoprecipitation (IP) with anti-Myc. (N and O) MKC cells were transfected with MDA5 plasmids and STING plasmids (N) or si-STING (O) for 24 h and then treated with SCR (MOI = 5) for 24 h. The expression of IFN-1 was determined by qPCR. (P and Q) MKC cells were transfected with MDA5 plasmids and si-STING for 24 h and then transfected with SCR-RNA for 24 h, then the IRF3 dimerization and IFN-1 levels were analyzed by native gel electrophoresis (P) and qPCR (Q), respectively. All data presented as the means $\pm$ SE from at least three independent triplicated experiments. **, $p < 0.01$; *, $p < 0.05$ versus the controls.

manner, to determine if the RD domain of *M. miiuy* MDA5 has similar functions, we generated truncated mutations called MDA5-ΔRD and Western blot analysis confirmed the expression of MDA5 and MDA5-ΔRD plasmids (**Figure 3B** [↗](#)). We subsequently noted that MDA5-ΔRD exhibited an inability to interact with SCR V RNA, which was supported by the loss of its immunological capability in inhibiting SCR V viral replication (**Figure 3C** [↗](#)-**3E** [↗](#)). We also examined the effects of MDA5 and its mutant on antiviral factors in both poly(I:C)-HMW-stimulated and SCR V-infected MKC cells. Results showed that only the overexpression of MDA5, not MDA5-ΔRD, significantly increased the expression of antiviral genes (**Figure 3F** [↗](#) and **3G** [↗](#)). In summary, these findings provided evidence that MDA5 may recognize 5'ppp-RNA through its RD domain.

2.4 MDA5 recognizes 5'ppp-RNA in *M. miiuy* and *G. gallus*

To provide specific evidence of MDA5 recognition of 5'ppp-RNA in *M. miiuy*, we synthesized 67 nt 5'ppp-SCR V from the SCR V virus and through in vitro transcription using T7 polymerase. As a positive control capable of binding to the mmiMDA5 receptor, 112 bp dsRNA from the MDA5 gene sequence was also generated. Gel analysis and nuclease sensitivity confirmed the generation of RNA products of the expected size (**Figure 4A** [↗](#)). The RNA pull-down experiment showed that 5'ppp-SCR V can be recognized by mmiMDA5, similar to 112 bp blunt end dsRNA (**Figure 4B** [↗](#)). Next, we transfected wild-type 5'ppp-SCR V and dephosphorylated mutant SCR V RNAs (5'-OH-SCR V and 5'pppGG-SCR V) into MKC cells to detect the dimer of IRF3, and found that the activation ability of 5'ppp-RNA on IRF3 depends on its triphosphate structure (**Figure 4C** [↗](#)). Furthermore, biotin-labeled 5'ppp-RNA captured mmiMDA5, while the biotin-labeled mutant types (5'-OH-SCR V and 5'pppGG-SCR V) did not (**Figure 4D** [↗](#)). Additionally, we purified recombinant MDA5 and MDA5-ΔRD of *M. miiuy* to conduct electrophoresis mobility shift assay (EMSA) (**Figure 4E** [↗](#)). When recombinant mmiMDA5 was subjected to EMSA using a biotin-labeled 5'ppp-SCR V probe, a clear complex was detected (**Figure 4F** [↗](#)). This complex formation was inhibited by adding an excess of cold 5'ppp-SCR V but not 5'-OH-SCR V and 5'pppGG-SCR V, demonstrating the direct interaction between mmiMDA5 and 5'ppp-SCR V (**Figure 4F** [↗](#)). However, 5'ppp-SCR V cannot bind to the MDA5-ΔRD (**Figure 4G** [↗](#)). Next, we investigated whether *G. gallus* MDA5 (ggaMDA5) shares the ability to recognize 5'ppp-RNA seen in fish. We synthesized 67 nt 5'ppp-VSV from the VSV virus and found that it is similar to 5'ppp-SCR V, binding to ggaMDA5 and activating IRF3 in a triphosphate-dependent manner (4H-4L). Overall, our findings demonstrated that MDA5 detection of RNA in two RIG-I deficient vertebrates (*M. miiuy* and *G. gallus*) is guided by the RNA 5'-triphosphate end and is disrupted when RNA is capped at the 5'-triphosphate end or dephosphorylated.

2.5 Increased m⁶A modification and expression of MDA5 upon SCR V infection

As stated in the previous results, MDA5 has evolved to possess immune recognition abilities that take over the role of RIG-I, significantly enhancing the host's ability to resist viral infections. However, evolution is a continuous battle between hosts and viruses, viruses often evolve complex escape mechanisms to evade immune surveillance and destruction, in which the m⁶A modification mechanism is an important way. To investigate the impact of SCR V infection on the overall methylation dynamics of fish hosts, we utilized methylated RNA immunoprecipitation sequencing (MeRIP-seq, GenBank accession number: PRJNA819945) to measure changes in m⁶A modification of host transcripts during SCR V infection (**Figure 5A** [↗](#)). From the results shown in **Figure 5B** [↗](#), we identified 3060 differentially m⁶A methylated peaks (p -value < 0.05) and 2082 differentially expressed genes (p -value < 0.05) between the non-infected and SCR V-infected group, indicating that infection significantly alters the m⁶A modification of specific host transcripts. It is worth noting that hyper-methylated m⁶A peaks were also observed in the MDA5 mRNA transcripts during SCR V infection (fold change >1.5, p < 0.05), and the expression level of MDA5 was higher in the infection group compared to the control group (p < 0.05). Further analysis of the m⁶A-seq data

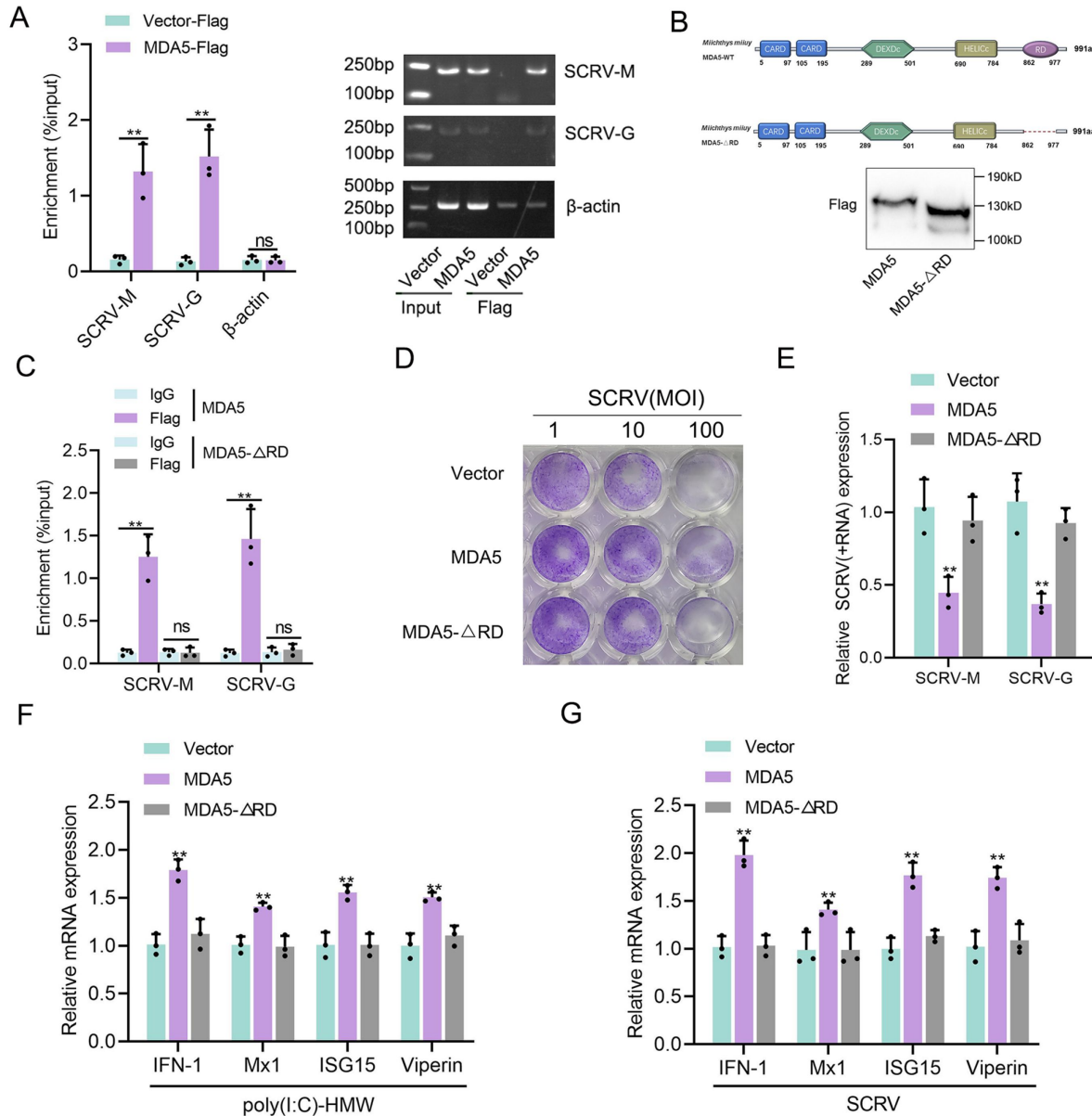


Figure 3.

RD domain is required for MDA5 to recognize SCR RNA.

(A) The association of MDA5 proteins with SCR. (B) Schematics and expression of MDA5 and truncated MDA5 (MDA5- Δ RD). (C) The association of MDA5 and MDA5- Δ RD proteins with SCR. (D) MKC cells transfected with pcDNA3.1 vector, MDA5, or MDA5- Δ RD plasmids for 24 h and then treated with SCR at the dose indicated for 48 h. Then, cell monolayers were fixed with 4% paraformaldehyde and stained with 1% crystal violet. (E) MKC cells were transfected with pcDNA3.1 vector, MDA5 or MDA5- Δ RD expression plasmids for 24 h, then infected with SCR for 24 h, then the qPCR analysis was conducted for SCR-V-M and SCR-V-G RNA levels. (F and G) Expression antiviral genes in poly(I:C)-HMW-stimulated (F) or SCR-infected (G) MKC cells transfected with pcDNA3.1 vector, MDA5, or MDA5- Δ RD expression plasmids. All data presented as the means $\pm$ SE from at least three independent triplicated experiments. **, $p < 0.01$; *, $p < 0.05$ versus the controls.

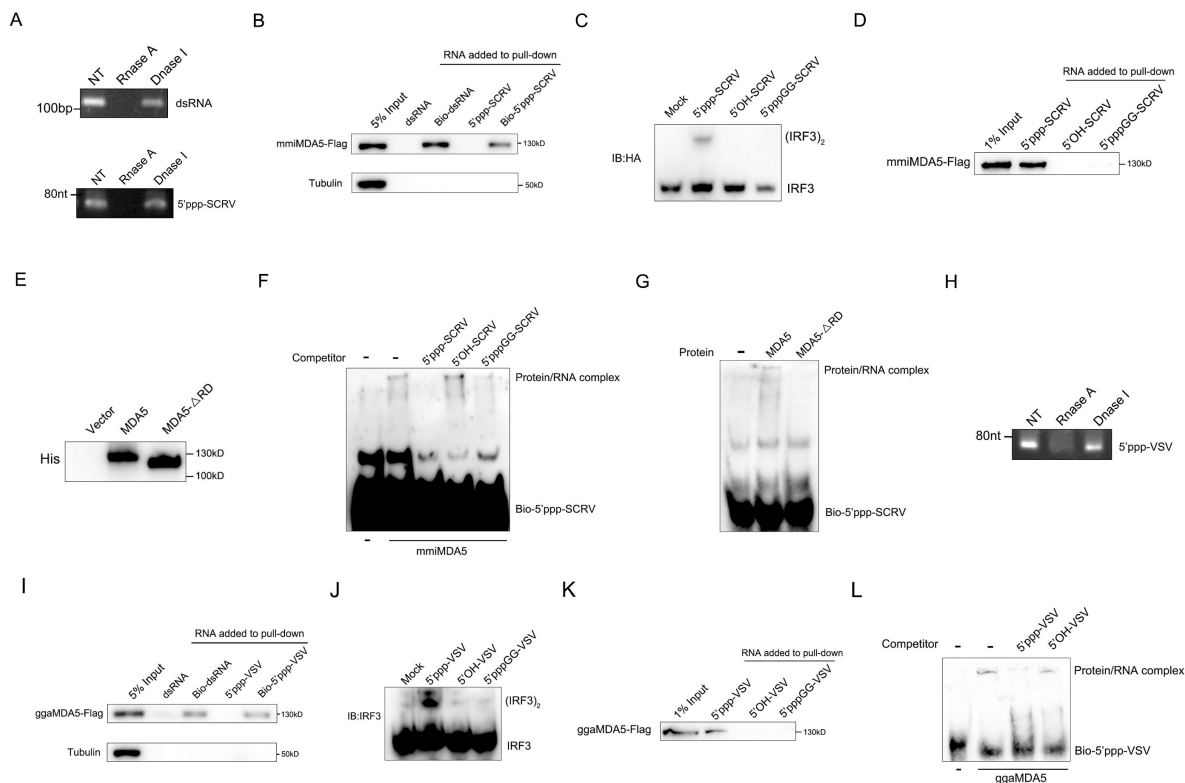


Figure 4.

MDA5 recognizes 5'ppp-RNA in *M. miiuy* and *G. gallus*.

(A) Schematic representation of 5'ppp-SCR and dsRNA. The product of in vitro transcription runs as a single product degraded by RNase I. (B) Pull-down of biotinylated or non-biotinylated dsRNA and 5'ppp-SCR by mmiMDA5. MKC cells were transfected with mmiMDA5-Flag plasmid, and the input and immunoprecipitated MDA5 proteins were analyzed by Western blot. (C) MKC cells were transfected with 5'ppp-SCR, 5'OH-SCR (5'ppp-SCR dephosphorylated by CIAP), or 5'pppGG-SCR (5'ppp-SCR capped by m7G cap analog), then IRF3 dimerization was analyzed by native gel electrophoresis. (D) The cytoplasmic fraction of MKC cells transfected with Flag-tagged mmiMDA5 was incubated with biotinylated 5'ppp-SCR, 5'OH-SCR, or 5'pppGG-SCR. RNA-protein complexes were pulled down using streptavidin affinity beads. Input and pull-down samples were analyzed by SDS-PAGE and immunoblotting using anti-Flag antibody. (E) Purity of recombinant mmiMDA5 and mmiMDA5-ΔRD were determined by SDS-PAGE and immunoblotting using anti-His antibody. (F) EMSA of 5'ppp-SCR with mmiMDA5. For binding competition, indicated non-biotinylated RNAs (50-fold molar excess over the probe) were included. (G) EMSA of 5'ppp-SCR with mmiMDA5 or mmiMDA5-ΔRD. (H) Schematic representation of 5'ppp-VSV. The product of in vitro transcription runs as a single product degraded by RNase I. (I) Pull-down of dsRNA and 5'ppp-VSV by ggaMDA5. DF-1 cells were transfected with ggaMDA5-Flag plasmids, and the input and immunoprecipitated MDA5 proteins were analyzed by Western blot. (J) DF-1 cells were transfected with 5'ppp-VSV, 5'OH-VSV (5'ppp-VSV dephosphorylated by CIAP), or 5'pppGG-VSV (5'ppp-VSV capped by m7G cap analog), then IRF3 dimerization was analyzed by native gel electrophoresis. (K) The cytoplasmic fraction of DF-1 cells transfected with Flag-tagged ggaMDA5 was incubated with biotinylated 5'ppp-VSV, 5'OH-VSV, or 5'pppGG-VSV. RNA-protein complexes were pulled down using streptavidin affinity beads. Input and pull-down samples were analyzed by SDS-PAGE and immunoblotting using anti-Flag antibody. (L) EMSA of 5'ppp-RNA with ggaMDA5. For binding competition, indicated non-biotinylated RNAs (50-fold molar excess over the probe) were included.

showed that MDA5 contained m⁶A sites in the exon1 region which were markedly higher in the infection group than that in the control group (Untreated VS SCR: $p = 0.01$) (**Figure 5C**). Next, we tested the concrete m⁶A modification sites of MDA5 mRNA. We used the SRAMP database to predict potential m⁶A sites in MDA5-exon1 ²⁴. Four potential sites were predicted and two specific primers were designed (**Figure 5D**), then m⁶A quantitative real-time PCR and semiquantitative PCR were performed. In line with the m⁶A-seq data, the results further confirmed that m⁶A sites exactly exist in MDA5 exon1 (**Figure 5E**). After being infected with SCR or poly(I:C)-HMW, the methylation level of MDA5 increased in MKC cells, further confirming the sequencing data (**Figure 5F**). Next, MDA5 was found to be upregulated in spleen and kidney tissues infected with SCR (**Figure 5G**). We further explored the mRNA and protein levels of MDA5 in MKC cells infected with SCR and poly(I:C)-HMW and found that their expression all first significantly increased and then returned to the normal level (**Figure 5H** and **5I**). Overall, our results suggested that both the m⁶A modification and expression levels of MDA5 are increased upon SCR infection.

2.6 m⁶A-modification weakens MDA5 mRNA stability and antiviral ability

SCR infection can increase the methylation level of MDA5, and this mechanism is expected to have a regulatory effect on MDA5. To investigate this, the regulatory effects of METTL3 and METTL14, the main components of the methyltransferase complex, on MDA5 were explored. Three siRNAs targeting METTL3 and METTL14 were designed, and si-METTL3-2 and si-METTL14-3 showed higher knockdown efficiency (**Figure 6A**). METTL3 and METTL14-overexpressing plasmids were also constructed, and the interaction between them confirmed their cooperative role in m⁶A modification (**Figure 6B**). Subsequently, the global m⁶A levels were measured in cells after knockdown or overexpression of METTL3 and METTL14. Consistent with expectations, the knockdown of METTL3 and METTL14 significantly decreased the global m⁶A level, while their overexpression significantly increased the overall methylation level (**Figure 6C**). To investigate their role in the process of virus infection, we used the virus plaque and qPCR methods to explore whether it could affect SCR replication. As shown in **Figure 6D** and **6E**, overexpression of METTL3 or METTL14 significantly enhances viral plaque formation and promotes SCR replication in SCR-infected cells. And as m⁶A writers, METTL3 and METTL14 can directly regulate the methylation level of MDA5 and induce MDA5 degradation by reducing its mRNA stability (**Figure 6F**–**6I**). We further inserted the MDA5-exon1 into pmirGLO and mVenus-C1 vector, and results showed that si-METTL3&14 can reduce the luciferase activity of pmirGLO-MDA5-exon1, and METTL3&14 could suppress the levels of GFP, further indicating the presence of methylation sites in MDA5-exon1 (Figure 6-figure supplement 1A and 1D). To investigate the potential sites on MDA5-exon1, we constructed mutant MDA5-exon1 reporter plasmids. For the mutant form of MDA5-exon1 reporter genes, four predicted adenosine bases in m⁶A consensus sequences (DRACH) were replaced by uracil, thus canceling m⁶A decoration (**Figure 5D** and Figure 6-figure supplement 1B). As expected, our results showed that the luciferase activity of wild-type pmirGLO-MDA5-exon1 reporter, but not of mutant pmirGLO-MDA5-exon1 reporter, was markedly increased upon METTL3&14 knockdown (Figure 6-figure supplement 1C). Furthermore, we found that the use of cycloleucine (CL, an amino acid analogue that can inhibit m⁶A modification) hindered the degradation of MDA5 by METTL3 and METTL14, indicating that these proteins regulate MDA5 expression through a methylation mechanism (**Figure 6J** and **6K**). Additionally, MDA5 expression was degraded by METTL3&14 in both SCR-infected and uninfected groups (**Figure 6L**). To investigate the impact of this mechanism on viral infection, we co-transfected METTL3&14 and MDA5 plasmids and found that METTL3&14 partially suppressed the antiviral immune benefit of MDA5 (**Figure 6M**–**6O**). Taken together, our data demonstrate that METTL3&14-mediated m⁶A methylation of MDA5 inhibits its mRNA stability and expression levels, thus suppressing the antiviral capabilities of MDA5.

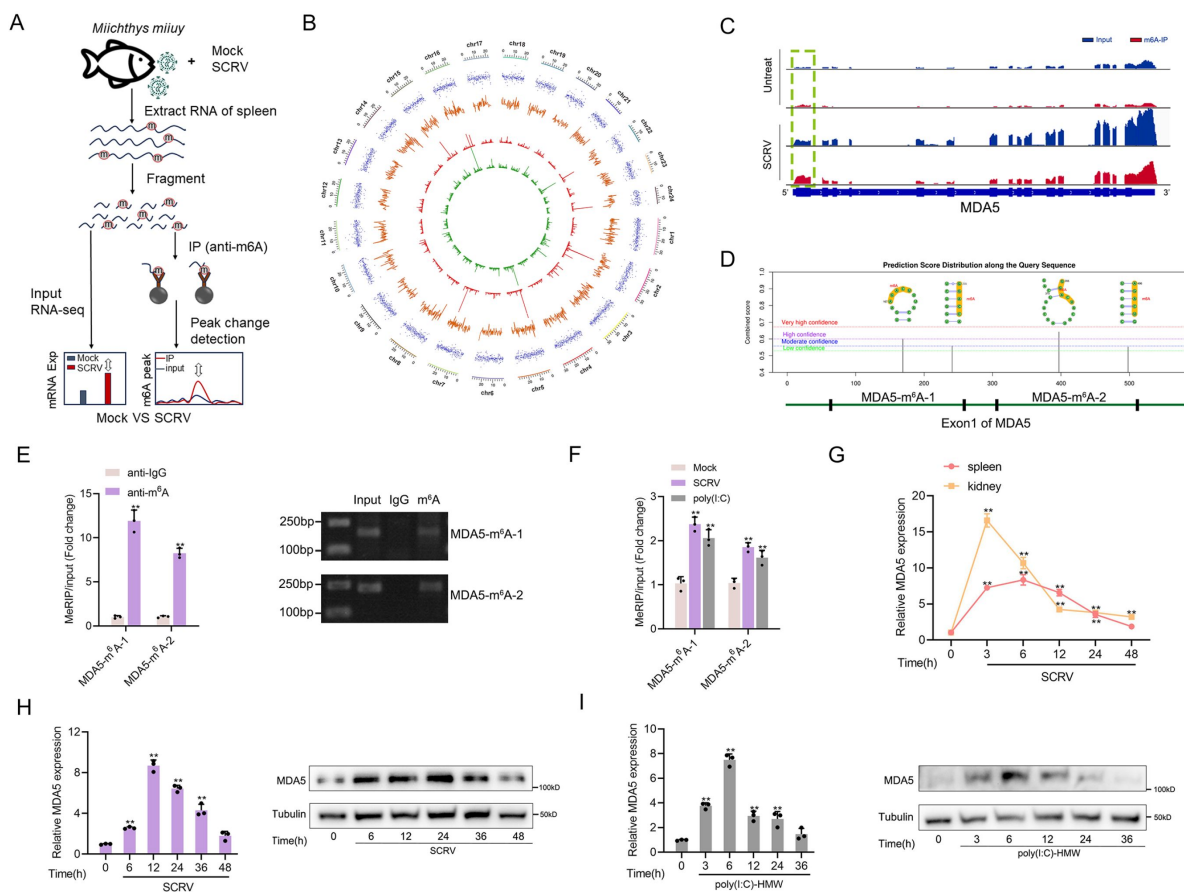


Figure 5.

Increased m⁶A modification and expression of MDA5 upon SCR infection.

(A) Schematic of the MeRIP-seq protocol used to identify differential m⁶A methylation following infection of *M. edulis* spleen tissues with SCR. (B) Circos plots showing differentially m⁶A-methylated peaks identified from normal and SCR-infected spleen tissues of *M. edulis*. The green and red lines on the innermost ring represented the m⁶A peaks identified from normal and SCR-infected spleen tissues of *M. edulis* respectively. The orange lines represent the log2 fold change values for each differentially m⁶A peak. The blue spots represent the fold change of gene expression. Chromosomes were shown on the outermost ring, the ruler on which represented the physical distance is in millions of bases (Mb). (C) The m⁶A abundance in MDA5 mRNA transcripts detected by MeRIP-seq, the m⁶A peak of MDA5 is circled in the green box. (D) The exon1 sequence of MDA5 was submitted to the SRAMP website, and then the predicted m⁶A site was displayed. Four predicted methylation sites located on the exon1 of MDA5 were marked by red box, then we designed two m⁶A specific primers (MDA5-m⁶A-1 and MDA5-m⁶A-2). (E) m⁶A abundance on MDA5 exon1 detected by MeRIP-qPCR and emiquantitative PCR in MKC cells. (F) MeRIP-qPCR analysis of relative m⁶A level of MDA5 in Mock, SCR-infected, and poly(I:C)-HMW-stimulated MKC cells. (G) mRNA levels of MDA5 in spleen and kidney tissues measured by qRT-PCR at indicated time after SCR infection (MOI = 5). (H and I) mRNA and protein levels of MDA5 in MKC cells measured by qRT-PCR and Western blotting at indicated time after SCR infection (MOI = 5) (H) and poly(I:C)-HMW stimulation (I). All data presented as the means ± SE from at least three independent triplicated experiments. **, *p* < 0.01 versus the controls.

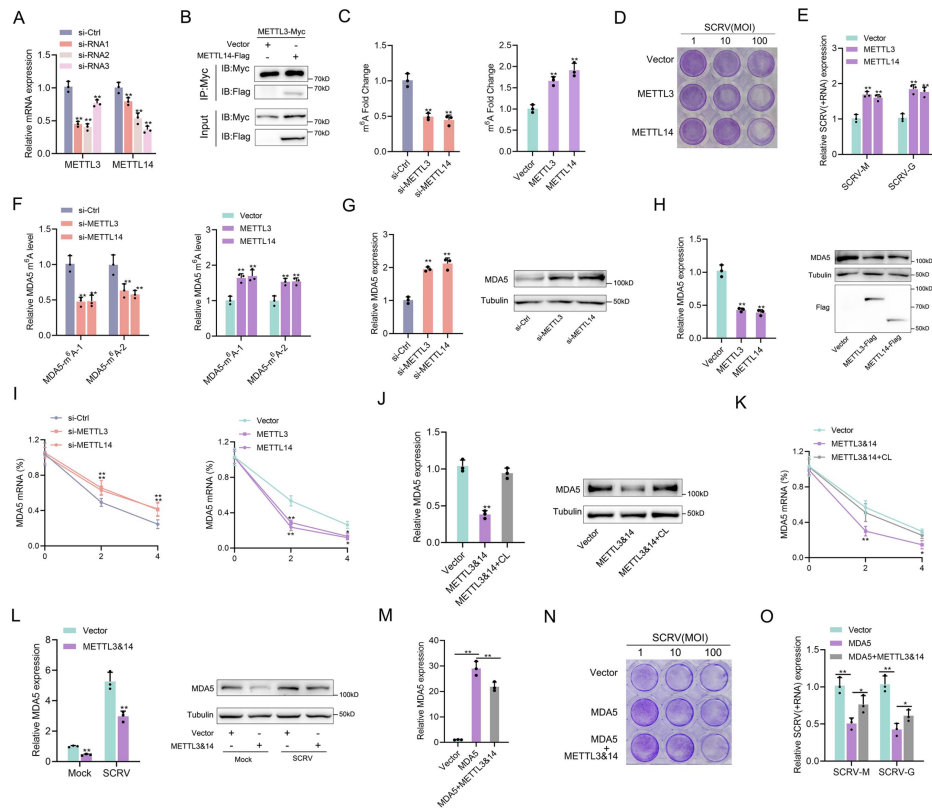


Figure 6.

m⁶A-modification weakens MDA5 mRNA stability and antiviral ability.

(A) Silence efficiency of si-METTL3 and si-METTL14 measured. Three siRNAs of METTL3 and METTL14 were transfected into MPC cells for 48 h respectively. (B) METTL3 immunoprecipitates with METTL14. EPC cells (1×10^7) were co-transfected with METTL3-Myc and METTL14-Flag expression plasmids for 24 h, followed by immunoprecipitation (IP) with anti-Myc. (C) METTL3 and METTL14 significantly increased the m⁶A content. MPCs were transfected with si-Ctrl or si-METTL3 or si-METTL14 and MKCs were transfected with vector or METTL3 or METTL14 plasmids for 48 h, then the m⁶A level was measured by colorimetry. (D) METTL3 and METTL14 overexpressed MKC cells seeded in 48-well plates overnight were treated with SCR virus at the dose indicated for 48 h. Then, cell monolayers were fixed with 4% paraformaldehyde and stained with 1% crystal violet. (E) MKC cells were transfected with pcDNA3.1 vector and METTL3 or METTL14 expression plasmid for 24 h, then infected with SCR virus (MOI = 5) for 24 h. The qPCR analysis was conducted for SCR-M and SCR-G RNA levels. (F) The m⁶A level alteration of MDA5 upon METTL3 or METTL14 knockdown or overexpression was examined by MeRIP-qPCR. MPC cells were transfected with si-Ctrl or si-METTL3 and MKC cells were transfected with vector or METTL3 or METTL14 plasmids for 48 h. (G and H) MPC cells were transfected with si-Ctrl, si-METTL3 or METTL14 and MKC cells were transfected with vector, METTL3, or METTL14 plasmids for 48 h, the expression of MDA5 was detected by qRT-PCR and Western blotting. (I) MPC cells were transfected with si-Ctrl, si-METTL3, or si-METTL14, and MKC cells were transfected with vector, METTL3, or METTL14 plasmids, then 5 μ g/ml actinomycin D was added to the cells for 0 h, 2 h, and 4 h. The half-life of MDA5 was analyzed by qRT-PCR. (J) MKC cells were transfected with vector or METTL3&14 plasmids for 24 h and then treated with Cycloleucine (CL) for 24 h in a final concentration of 20 mM. The expression of MDA5 was detected by qRT-PCR and Western blotting. (K) MKC cells were transfected with vector or METTL3&14 plasmids for 24 h and then treated with CL for 24 h at 20 mM, then 5 μ g/ml actinomycin D was added to the cells for 0 h, 2 h, and 4 h. The half-life of MDA5 was analyzed by qRT-PCR. (L) MKC cells were transfected with vector or METTL3&14 plasmids for 24 h and stimulated with SCR virus (MOI = 5) for 24 h, then the expression of MDA5 was detected by qRT-PCR and Western blotting. (M) MKC cells seeded in 48-well plates overnight were transfected with MDA5 or MDA5+METTL3&14 plasmids for 48 h, then the expression of MDA5 was detected by qRT-PCR. (N) MKC cells seeded in 48-well plates overnight were transfected with MDA5 or MDA5+METTL3&14 plasmids were treated with SCR virus at the dose indicated for 48 h. Then, cell monolayers were fixed with 4% paraformaldehyde and stained with 1% crystal violet. (O) MKC cells were transfected with MDA5 or MDA5 + METTL3&14 plasmids for 24 h, then infected with SCR virus (MOI = 5) for 24 h. The qPCR analysis was conducted for SCR-M and SCR-G RNA levels. All data presented as the means $\pm$ SE from at least three independent triplicated experiments. **, $p < 0.01$; *, $p < 0.05$ versus the controls.

2.7 Detailed m⁶A regulatory mechanism of MDA5

The m⁶A mechanism is jointly regulated by writers, erasers, and readers. The m⁶A modification of mRNA recruits m⁶A-binding proteins (readers) which have the function of recognizing m⁶A sites and accelerating mRNA decay. In this study, we hypothesized that YTHDF readers (YTHDF1, YTHDF2, and YTHDF3) may recognize m⁶A-modified MDA5 and destabilize it. We examined the binding ability of YTHDF proteins to MDA5 mRNA and found that YTHDF2 and YTHDF3 exhibited stronger binding compared to YTHDF1 (**Figure 7A** [↗](#)). We then designed two siRNAs targeting YTHDF1, YTHDF2, and YTHDF3 respectively, and selected si-YTHDF1-1, si-YTHDF2-si-2, si-YTHDF3-2 with better knockdown effects for subsequent experiments (**Figure 7B** [↗](#)). Results showed that the knockdown or overexpression of YTHDF2 and YTHDF3 led to the increase or decrease of the expression levels of MDA5, whereas YTHDF1 did not appear to influence MDA5 expression (**Figure 7C** [↗](#), **7D** [↗](#), Figure 7-figure supplement 1A and 1C). In addition, overexpression of YTHDF2&3 can increase the luciferase activity of wild-type pmirGLO-MDA5-exon1 reporter, but not of mutant pmirGLO-MDA5-exon1 reporter (Figure 7-figure supplement 1B). We then found that si-METTL3&14 can counteract the negative impact of YTHDF2&3 on MDA5 mRNA, protein, and stability (**Figure 7E** [↗](#) and **7F** [↗](#)). Since the m⁶A mechanism is reversible, we then studied the regulation of MDA5 by two major erasers. Results showed that FTO, not ALKBH5, can upregulate MDA5 expression (**Figure 7G** [↗](#)). Moreover, the overexpression of FTO can reverse the inhibitory effect of YTHDF2&3 on MDA5 (**Figure 7H** [↗](#) and **7I** [↗](#)). Taken together, these data suggested that through METTL3&14-m⁶A-YTHDF2&3 regulatory network, m⁶A can mediate the degradation of MDA5, and this process can be countered by FTO (**Figure 7J** [↗](#)).

3. Discussion

In this study, we have identified multiple independent loss events of RIG-I throughout vertebrate evolution. Specifically, we have observed at least three independent loss events in mammals, birds, and fish, respectively. Regarding the loss of RIG-I in avian species, previous studies have provided more comprehensive and accurate information than ours, indicating that birds have experienced more than 16 instances of RIG-I loss¹⁰ [↗](#). However, it should be noted that we have not found the loss of RIG-I in reptiles and amphibians (not displayed). But this does not imply that there has been no loss event among them, but rather, there is insufficient information regarding the genomes of reptiles and amphibians to draw the conclusions. As the abundance of genomic data continues to grow, there will be an increasing amount of evidence available to verify the presence or absence of RIG-I in these taxonomic groups. Additionally, the presence of RIG-I in current species does not guarantee that RIG-I loss has not occurred previously. For instance, our research has indicated that zebrafish may have experienced RIG-I loss. However, prior to the loss event, the zebrafish RIG-I has undergone repetitive events. These findings collectively highlight that the loss of RIG-I is not a random occurrence, but rather bears significant biological and evolutionary implications that deserve further investigation.

The loss of RIG-I might be expected to have a great impact on immune response, and the evolutionary loss of it raises the question of a possible compensation of its function by other gene products. RIG-I deficiency instances in mammals were scarcer compared to those in fish and birds. Previous research has revealed that in *T. belangeri*, concurrent with the loss of RIG-I, both MDA5 and LGP2 have undergone notable positive selection, and MDA5 or MDA5/LGP2 could sense Sendai virus (an RNA virus posed as a RIG-I agonist) for inducing type I IFN. This proposed an immune substitution model for RIG-I deficiency in mammals, though direct evidence of MDA5/LGP2 binding to 5'ppp-RNA was lacking. In this study, we characterized the consequences resulting from the loss of RIG-I in the *M. miiuy* and *G. gallus* (two representative species of vertebrates lacking RIG-I) and provided direct functional evidence for the alternative immune recognition function of

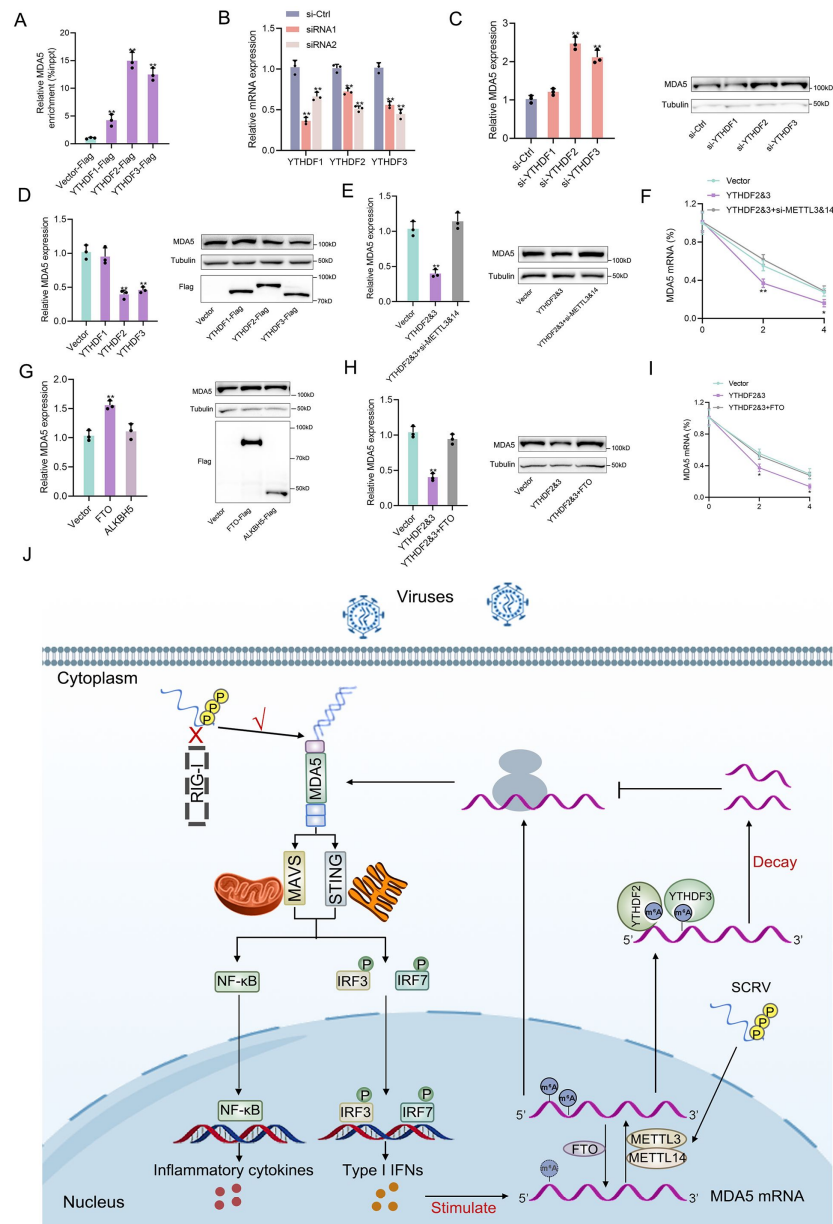


Figure 7.

Detailed m⁶A regulatory mechanism of MDA5.

(A) The binding relationship between YTHDF1, YTHDF2, or YTHDF3 and MDA5 mRNA was validated using a RIP assay. MKC cells were transfected with YTHDF1-Flag, YTHDF2-Flag, YTHDF3-Flag, or pcDNA3.1-Flag for 48 h. (B) SiRNA silencing effect test of YTHDF1, YTHDF2, and YTHDF3. MPC cells were transfected with si-YTHDF1, si-YTHDF2, or si-YTHDF3 for 48 h. (C and D) MPC cells were transfected with si-YTHDF1, si-YTHDF2, si-YTHDF3 or si-Ctrl (C), and MKC cells were transfected with YTHDF1, YTHDF2, YTHDF3 or vector for 48 h (D), then the expression of MDA5 was detected by qRT-PCR and Western blotting. (E) Relative mRNA and protein levels of MDA5 in MKC cells after co-transfected with YTHDF2&3 plasmids and si-METTL3&14 by qPCR and western blot assays. (F) MKC cells were transfected with YTHDF2&3 plasmids and si-METTL3&14, then 5 μ g/ml actinomycin D was added to the cells for 0 h, 2 h, and 4 h. The half-life of MDA5 was analyzed by qRT-PCR. (G) MKC cells were transfected with vector, FTO, ALKBH5 plasmids for 48 h, then the expression of MDA5 was detected by qRT-PCR and Western blotting. (H) Relative mRNA and protein levels of MDA5 in MKC cells after co-transfected with YTHDF2&3 plasmids and FTO plasmids by qPCR and western blotting. (I) MKC cells were transfected with YTHDF2&3 plasmids and FTO plasmids, then 5 μ g/ml actinomycin D was added to the cells for 0 h, 2 h, and 4 h. The half-life of MDA5 was analyzed by qRT-PCR. (J) Schematic diagram of arms race between MDA5 and 5'ppp-RNA virus in *M. miiuyi*.

MDA5. SCR.V and VSV belongs to the class of rhabdovirus, which is a negative-strand virus with a non-segmented genome and initiate both replication and transcription de novo leading to 5'-triphosphate RNA in the cytosol, was expected to trigger an IFN response without the need for replication and presumed dsRNA formation⁴. Then, we synthesized 5'ppp-RNA probes from the SCR.V and VSV genome sequences and purified recombinant MDA5 of *M. miiuy* and *G. gallus*. Results demonstrated that MDA5 detect the 5'ppp-RNA in *M. miiuy* and *G. gallus* and is disrupted when RNA is capped at the 5'-triphosphate end or dephosphorylated. Furthermore, the recognition of RNA by MDA5 depends on its RD domain, which not only indicates that the RD domain of MDA5 has evolved functions similar to RD domain of RIG-I, but also excludes the possibility of MDA5 binding to 5'ppp-RNA through intermediate ligands. Furthermore, in vertebrates that possess RIG-I, STING exclusively binds to RIG-I rather than MDA5. However, similar to the interaction observed between MDA5 and STING in tree shrews and chickens that lacking RIG-I^{8, 25}, the MDA5 of *M. miiuy* can also interact with STING protein, indicating that in vertebrates lacking RIG-I, MDA5 may utilize STING to facilitate signal transduction in the antiviral response. Finally, by linking the immune replacement ability evolved by MDA5 with the common RIG-I loss event in vertebrates, we propose a straightforward explanation for this evolutionary event: In the lengthy process of vertebrate evolution, MDA5 gradually acquired new functionalities to effectively detect viruses such as 5'ppp-RNAs, which were initially recognized by RIG-I. As a result, there was functional redundancy of RIG-I, which eventually led to its gradual disappearance from the vertebrate genome. This hypothesis is supported by two viewpoints: first, MDA5 and RIG-I can both recognize overlapping viruses²⁶; secondly, in fish without RIG-I deficiency, MDA5 has stronger antiviral ability compared to RIG-I²⁷.

Viruses are constantly co-evolving with hosts. Our available data suggested that the vertebrate has evolved MDA5 with alternative functions to recognize and resist 5'ppp-RNA virus in the absence of RIG-I. As a response, virus has also evolved to use the m⁶A mechanism to evade the host immune response. Previous study indicated that m⁶A modification plays a crucial role in maintaining immune homeostasis in the body²⁸. During the infection process, the virus may disrupt the m⁶A level and disrupt the immune balance, thereby facilitating its own replication and invasion. Viruses may utilize the m⁶A mechanism in two ways to evade immunity. On the one hand, many studies suggested a role for m⁶A in shielding RNA species from detection by PRRs. There are numerous viruses recognized in RIG-I dependent manner, but m⁶A modification increases the possibility of immune evasion. For example, the loss of m⁶A site or removing m⁶A from human metapneumovirus may weaken its immune evasion ability, prevent it from evading detection by RIG-I, and result in higher levels of IFN release²⁰. Moreover, it has been reported that the RNA containing m⁶A modifications binds to RIG-I poorly, and it failed to trigger the RIG-I conformational changes associated with downstream immune signals²⁹. On the other hand, m⁶A-mediated viral strategies for inhibiting IFN induction pathways appear to exist. For instance, following infection by human cytomegalovirus (HCMV), the IFN- β transcript was modified by m⁶A, and this methylation decreased its half-life, suggesting that HCMV can exploit this control of IFN- β expression to facilitate its replication by increasing m⁶A modification of the IFN- β transcript and thus decreasing its production³⁰. In another study, the m⁶A reader YTHDF3 could restrain the type I IFN response and ISG expression by involving enhancement of FOXO3 translation in a m⁶A-independent mechanism, which acts as a negative regulator of immunity¹⁹. Moreover, previous study reports that IFN- α and IFN- β mRNA encoding cytokines that promote IFN response are modified by m⁶A, and the loss of METTL3 and YTHDF2 could increase the IFN expression and ISG activity³¹. In summary, viruses can utilize m⁶A to evade the immune response by shielding the recognition of RIG-I or reducing the host's immune reaction, thus counteracting the host's defense mechanisms. In our previous study, we discovered that SCR.V infection can not only increase overall cellular methylation levels but also upregulate the expression of METTL3, thereby suggesting its potential to regulate host gene expression through the m⁶A mechanism³². In this study, we found that SCR.V virus infection alters m⁶A modification of specific cellular transcripts, including MDA5. MDA5 plays a crucial role as a receptor in recognizing the SCR.V virus and controlling IFN activation. The aberrant m⁶A levels of MDA5 subsequently affect the expression of

antiviral genes, thereby compromising its ability to mount an effective antiviral immune response. It is plausible that SCR_V also impacts the production of antiviral genes, such as IFN-1 or other cytokines, by manipulating m⁶A modification on the transcripts of cytokines or molecules involved in their production. This interesting aspect warrants further investigation in future research.

In this study, we demonstrated a co-evolutionary arms race between 5'ppp-RNA virus and virus receptors. Specifically, we conducted an evolutionary analysis and provided functional evidence to confirm that MDA5 of *M. miiuy* and *G. gallus* has acquired an additional function of sensing 5'ppp-RNA, thus compensating for the loss of RIG-I. Additionally, we found that SCR_V infection can regulate the m⁶A level of MDA5 in the *M. miiuy*, leading to its degradation and subsequently affecting the immune response for immune evasion. Fish and birds, as the most abundant aquatic and terrestrial vertebrates respectively, are also the two lineages with the most frequent occurrence of RIG-I deficiency, the investigation of host-pathogen interactions of them offers valuable insights into the ecological and evolutionary factors that contribute to the diversity of the immune system.

4. Materials and methods

4.1 Ethics statement

All animal experimental procedures were performed in accordance with the National Institutes of Health's Guide for the Care and Use of Laboratory Animals, and the experimental protocols were approved by the Research Ethics Committee of Shanghai Ocean University (No. SHOU-DW-2018-047).

4.2 Fish and challenge

M. miiuy (~50 g) was obtained from Zhoushan Fisheries Research Institute, Zhejiang Province, China. Fish was acclimated in aerated seawater tanks at 25°C for six weeks before experiments. The experimental procedure for SCR_V infection was performed as described³³. Briefly, In the SCR_V injection group, fish was challenged with 200 µl SCR_V at a multiplicity of infection (MOI) of 5 through intraperitoneal. As a comparison, 200 µl of physiological saline was used to challenge the individuals. Afterwards, fishes were respectively sacrificed at different time point and the kidney and spleen tissues were collected for RNA extraction.

4.3 Cell culture and treatment

M. miiuy kidney cells (MKCs) and spleen cells (MPCs) are cell lines derived from the kidney and spleen tissues of *M. miiuy*³⁴, with passages ranging from 20 to 40 times. MKC and MPC cells were cultured in L-15 medium (HyClone) supplemented with 15% fetal bovine serum (FBS; Gibco), 100 U/ml penicillin, and 100 µg/ml streptomycin at 26°C. Fish EPC cells (Epithelioma papulosum cyprini cells) were maintained in medium 199 (Invitrogen) supplemented with 10% FBS, 100 U/ml penicillin, and 100 mg/ml streptomycin at 26°C in 5% CO₂. HEK293T cells were cultured in DMEM (HyClone) supplemented with 10% FBS, 100 U/ml penicillin, and 100 mg/ml streptomycin at 37°C in 5% CO₂. Chicken embryo fibroblast cells (DF-1) were cultured in DMEM (HyClone) supplemented with 10% FBS, 100 U/ml penicillin, and 100 mg/ml streptomycin at 39°C in 5% CO₂. For stimulation experiments, cells were challenged with SCR_V at a multiplicity of infection (MOI) of 5 and harvested at different times for RNA extraction.

4.4 Oligonucleotides

The sequence of the 62 nt 5'ppp-RNA was derived from the 5' and 3'UTRs of the VSV genome as previously described³⁷, and the sequence of 112 bp dsRNA was taken from the first 100 bp of the MDA5 gene of *miiuy* croaker flanked by 5'-gggaga and tctccc-3'. 67 nt 5'ppp-SCR or 5'ppp-VSV and 112 bp dsRNA were synthesized by in vitro transcription using the Ribo RNAmix-T7 kit (RiboBio). Biotin-labeled RNAs were transcribed in vitro using a Ribo RNAmix-T7 Biotin Labeling Transcription Kit (RiboBio). For generating duplexes, RNA oligonucleotides were mixed in hybridization buffer (20 mM Tris-HCl [pH 8.0], 1.5 mM MgCl₂, and 1.5 mM DTT), boiled for 1 min, and incubated at 37°C for 1 h. For removal of 5'-triphosphates, 20 mg of RNA was treated with 20 U Calf Intestinal Alkaline Phosphatase (CIAP, Invitrogen) for 2 h at 37°C in the presence of RNase inhibitor (Beyotime) and extracted with phenolchloroform. RNA molecules with a terminal 5' m7G cap were synthesized by the incorporation of m7G cap analogue (Yeason) in the transcription reactions. RNA was analysed on a denaturing 17% polyacrylamide, 7 M urea gel following digestion with 50 ng/μl of RNase A (Beyotime) or 100 mU/μl of DNase I (Beyotime) for 30 min.

4.5 Plasmids construction

To construct Flag or Myc-tagged MDA5 (GeneBank: PP179381.1), MAVS, STING, METTL3, METTL14, FTO, ALKBH5, YTHDF1, YTHDF2, and YTHDF3 expression plasmids, their CDS sequences of the *M. miiuy* were amplified by PCR and then ligated into a pcDNA3.1 vector (Invitrogen), respectively. Then, MDA5-ARD were generated by PCR on the basis of MDA5 plasmid. To construct MDA5-exon1 reporter vector, the exon1 region of *M. miiuy* MDA5 gene was amplified using PCR with the gene-specific primers and ligated into a pmir-GLO luciferase reporter vector (Promega). Meanwhile, *M. miiuy* MDA5-exon1 were inserted into the mVenus-C1 (Invitrogen), which included the sequence of enhanced GFP. For mutant reporter vectors (MDA5-exon1-mut), adenosine (A) in the predicted m⁶A motif was replaced by an uracil (U) by using the Mut Express II Fast Mutagenesis Kit V2 (Vazyme) with specific primers). To construct Flag-tagged MDA5 expression plasmid of *G. gallus*, the CDS sequence of MDA5 was amplified by PCR and then ligated into a pcDNA3.1 vector (Invitrogen). The correct construction of the recombinant plasmids was verified through Sanger sequencing and extracted using an Endotoxin-Free Plasmid DNA Miniprep Kit (Tiangen, China) before using plasmids. The primers were listed in Supplementary file 1.

4.6 Protein Purification and Analysis

For protein purification, MDA5 plasmids with 6× His tag was constructed based on pcDNA3.1. (His6)-Flag-tagged MDA5 of *M. miiuy* and *G. gallus* were transiently overexpressed in 293T cells and lysed in a binding buffer (500 mM NaCl, 20 mM Tris/HCl [pH 7.4], 20 mM Imidazole, and 1% Triton X-100) including protease inhibitor cocktail (Roche). The lysate was incubated over night at 4°C with anti-His beads (Solarbio). Anti-His beads were washed subsequently with washing buffer (500 mM NaCl, 20 mM Tris/HCl [pH 7.4], 20 mM Imidazole, and 1% Triton X-100). MDA5 proteins were eluted by an addition of elution buffer (500 mM NaCl, 20 mM Tris/HCl [pH 7.4], 500 mM Imidazole, and 1% Triton X-100) to the beads. Purity of recombinant MDA5 was determined by SDS-PAGE separation.

4.7 RNA interference

The MDA5-specific small interfering RNA (si-MDA5-1 and si-MDA5-2) sequences were 5'-CGGACUACAUGCAGCGUAATT-3' and 5'-GACCAUGAGAUUUCUAUGTT-3', respectively. The LGP2-specific small interfering RNA (si-LGP2) sequence was 5'-GGUUCACCUUGUAGAGCAATT-3'. The STING-specific small interfering RNA (si-STING) sequence was 5'-ACGACAGCAUCCAUUUCUATT-3'. The METTL3-specific small interfering RNA (si-METTL3-1, si-METTL3-2, and si-METTL3-3) sequences were 5'-CAUCACAAACGAACUAACTT-3', 5'-AACGUGGGCAAACUCUUUUTT-3' and 5'-GAGAUUCUGGAACUACUATT-3', respectively. The METTL14-specific small interfering RNA (si-METTL14-1, si-METTL14-2, and si-METTL14-3) sequences were 5'-GCUGGAAUUCGAGGAGAUATT-3',

5'-CCGTACGAAGAGGTGTACATT-3' and 5'-GAGCCUCCUUGGAAGAAUTT-3', respectively. The YTHDF1-specific small interfering RNA (si-YTHDF1-1 and si-YTHDF1-2) sequences were 5'-AAAGACUUUGACUGGAACUTT-3' and 5'-CAGUCGAUCAGAGACCUAATT-3', respectively. The YTHDF2-specific small interfering RNA (si-YTHDF2-1 and si-YTHDF2-2) sequences were 5'-GCAGGGUGUUUAUCAUCAATT-3', 5'-CUAUGCUCACAGCUAAUUTT-3', respectively. The YTHDF3-specific small interfering RNA (si-YTHDF3-1 and si-YTHDF3-2) sequences were 5'-GGUGGACUACAAUGCCUAUTT-3' and 5'-UCUACAGUACAGCUAUGGTT-3', respectively. The scrambled control RNA sequences were 5' – UUCUCCGAACGUGUCACGUTT -3'. They were purchased from GenePharma (Shanghai, China).

4.8 Cells transfection

Transient transfection of cells with siRNA was performed in 24-well plates using Lipofectamine RNAiMAX (Invitrogen), and cells were transfected with DNA plasmids was performed using Lipofectamine 3000 (Invitrogen) according to the manufacturer's instructions. For functional analyses, the overexpression plasmid (500 ng per well) or control vector (500 ng per well) and siRNA (100 nM) were transfected into cells in culture medium and then harvested for further detection.

4.9 RNA extraction and quantitative real-time PCR

Total RNA was extracted using the TRIzol Reagent (Invitrogen) and the cDNA was synthesized using the FastQuant RT Kit (Tiangen) which contains DNase treatment of RNA to eliminate genomic contamination, following the manufacturer's instructions. The expression profiles of each gene were conducted by using the SYBR Premix Ex Taq™ (Takara), as previously described. The quantitative real-time PCR was conducted in an Applied Biosystems® QuantStudio 3 (Thermo Fisher Scientific). β -actin was used as internal controls for mRNA respectively. Primer sequences are listed in Supplementary file 1.

4.10 Western blotting

Cellular lysates were generated by using 1×SDS-PAGE loading buffer. Proteins were extracted from cells and measured with the BCA Protein Assay kit (Vazyme), then subjected to SDS-PAGE (10%) gel and transferred to PVDF (Millipore) membranes by semidry blotting (Bio-Rad Trans Blot Turbo System). The membranes were blocked with 5% BSA. Protein was blotted with different antibodies. The antibody against MDA5 was diluted at 1: 500 (Beyotime, Cat# AF7164); The antibody against IRF3 was diluted at 1: 500 (Boster, Cat# BA4351-2); anti-Flag, anti-Myc, anti-HA and anti-Tubulin monoclonal antibody were diluted at 1: 1,000 (Beyotime); and HRP-conjugated anti-rabbit IgG or anti-mouse IgG (Abbkine) at 1: 5,000. The results were the representative of three independent experiments. The immunoreactive proteins were detected by using WesternBright™ ECL (Advansta). The digital imaging was performed with a cold CCD camera.

4.11 Dual-luciferase reporter assays

To determine the functional regulation of MDA5, MKC cells were co-transfected MDA5 expression plasmid, together with NF- κ B, IRF3, IRF7, or IFN-1 luciferase reporter genes, and the pRL-TK *Renilla* luciferase reporter plasmid. After 48 h transfection, the cells were collected and lysed for reporter activity using the dual-luciferase reporter assay system (Promega). For the detection of m⁶A site, the wild or mutant of MDA5-exon1 luciferase reporter was co-transfected si-METTL3, si-METTL14, si-YTHDF1, si-YTHDF2, si-YTHDF3 or si-Ctrl into MKC cells. After 48 h transfection, the cells were collected and lysed for the reporter luciferase activities measured by using a dual-luciferase reporter assay system (Promega). All the luciferase activity values were gained against the *Renilla* luciferase control. For each experiment, three independent experiments were conducted, and each experiment was done in triplicate.

4.12 RNA binding protein immunoprecipitation (RIP)

To identify whether MDA5 or MDA5- Δ RD can directly bind to SCRv viral RNA, MKC cells ($\sim 2.0 \times 10^7$) were transfected with pcDNA3.1-Flag, pcDNA3.1-MDA5-Flag or pcDNA3.1-MDA5- Δ RD-Flag and then infected with SCRv at an MOI of 5 for 24 h. In addition, to identify whether YTHDF1, YTHDF2, and YTHDF3 can directly bind to MDA5, MKC cells were co-transfected with pcDNA3.1-Flag, pcDNA3.1-YTHDF1/2/3-Flag. For RNA immunoprecipitation (RIP) assays, MKC cells were harvested after 48 h transfection, and RIP assays were carried out with Magna RIP RNA-Binding Protein Immunoprecipitation Kit (Millipore) and anti-Flag antibody (Abcam) following the manufacturer's protocol. Then, the expression of target gene SCRv-M, SCRv-G, β -actin, or MDA5 were detected by quantitative real-time qRT-PCR analysis.

4.13 RNA Pull-Down assay

Cytoplasmic extracts were prepared from 2×10^7 MKC cells transfected with mmiMDA5-Flag (MDA5 of *miuiy* croaker) plasmids and 5×10^7 DF1 cells transfected with gga-MDA5-Flag (MDA5 of chicken) plasmids. The extracts were incubated with biotinylated 5'ppp-RNA or dsRNA and subjected to pull-down with streptavidin agarose beads (Genesee), followed by subsequent Coomassie bluestaining or SDS-PAGE analysis and immunoblotting with anti-Flag antibody (Sigma).

4.14 EMSA assay

Recombinant MDA5 proteins were mixed with biotin-labeled oligonucleotides in a reaction mixture (10 μ l: 20 mM Tris-HCl pH 8.0, 1.5 mM $MgCl_2$, 1.5 mM DTT). After incubation at room temperature for 15 min, the reaction mixtures were conducted on 6% non-denaturing polyacrylamide gels and the EMSA was performed using Chemiluminescent EMSA Kit (Beyotime).

4.15 MeRIP-seq assays and analysis

Total RNA was isolated using TRIzol reagent. m^6A immunoprecipitation and library preparation were performed following the manufacturer's protocol. Poly (A) RNA is purified from 50 μ g total RNA using Dynabeads Oligo (dT) (Thermo Fisher) using two rounds of purification. Then the poly(A) RNA was fragmented into small pieces using Magnesium RNA Fragmentation Module (NEB) under 86°C for 7 min. Then the cleaved RNA fragments were incubated for 2 h at 4°C with m^6A -specific antibody (Synaptic Systems) in IP buffer (50 mM Tris-HCl, 750 mM NaCl and 0.5% Igepal CA-630). Then the IP RNA was reverse-transcribed to create the cDNA by SuperScript™ II Reverse Transcriptase (Invitrogen). The average insert size for the final cDNA library was 300 ± 50 bp. For sequencing data analysis, we used HISAT2 to map reads to the reference genome of *M. miuiy*. Mapped reads of IP and input libraries were provided for R package exomePeak, which identifies m^6A peaks with bed or bigwig format that can be adapted for visualization on the IGV software. MEME and HOMER were used for de novo and known motif finding followed by localization of the motif with respect to peak summit. Called peaks were annotated by intersection with gene architecture using R package ChIPseeker. The differentially expressed mRNAs were selected with $\log_2$ (fold change) > 1 or $\log_2$ (fold change) < -1 and p -value < 0.05 by R package edgeR. The circus map of m^6A peaks in untreated and SCRv-infected spleen tissues of *M. miuiy* was showed by TBtools³⁵.

4.16 MeRIP-qPCR

Total RNA was extracted using TRIzol reagent (Invitrogen). Methylated m^6A RNA immunoprecipitation (Me-RIP) was performed according to the protocol of EpiQuik™ CUT&RUN m^6A RNA Enrichment Kit (Epigentek). qPCR analysis of the methylated RNA was performed to detect methylated MDA5 mRNA levels.

4.17 Total m⁶A Quantification Assay

Total RNA was extracted by the TRIzol Reagent (Invitrogen). Then, an EpiQuik m⁶A RNA Methylation Quantification Kit (EpiGentek) was used to detect the total m⁶A level according to the manufacturer's protocol. Briefly, positive control (PC), negative control (NC), and 200 ng isolated mRNA were added to each well with the capture antibody. Next, the RNA was incubated with m⁶A antibody in wells, and m⁶A content was quantified at a wavelength of 450 nm.

4.18 RNA stability assays

Cells were treated with actinomycin D (5 µg/mL) and then collected at different time points. RNA was extracted by TRIzol reagent (Invitrogen), and the mRNA levels were measured using qRT-PCR.

4.19 Database mining and sequence analysis

The species tree is constructed by submitting species names to the NCBI (<https://www.ncbi.nlm.nih.gov/Taxonomy/CommonTree/wwwcmt.cgi>). In order to determine whether there is RIG-I in vertebrates, we use *H. sapiens* and *D. rerio* RIG-I protein sequences to TBLASTN the whole genome sequence of the species on the ensemble website (<http://www.ensembl.org/>). In addition, in order to prevent the possibility of incomplete species genome, we also used RIG-I protein sequences of *H. sapiens* and *D. rerio* to compare the transcriptome and genome sequenced of the designated species on NCBI website (<http://www.ncbi.nlm.nih.gov/Genbank/>). For further gene synteny analysis, RIG-I of *H. sapiens*, *C. milii*, *D. rerio* were used as anchor sites. In order to identify the MDA5 and LGP2 from *M. miiuy*, we used the homologues of zebrafish reported previously as queries to seek for the transcriptome³⁶ and a chromosome-scale genome database³⁷ of *M. miiuy* by using local TBLASTN and BLASTN programs, then the protein structures were predicted by SMART website (<http://smart.embl-heidelberg.de/>).

4.20 Statistical analysis

Data are expressed as the mean ± SE from at least three independent triplicated experiments. A two-sided Student's t test was used to evaluate the data. The relative gene expression data was acquired using the 2^{-ΔΔCT} method, and comparisons between groups were analyzed by one-way analysis of variance (ANOVA) followed by Duncan's multiple comparison tests. A value of *p* < 0.05 was considered significant.

Acknowledgements

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Author contributions

S.G. designed and performed experiments, analyzed the data, performed bioinformatics analysis, and wrote the manuscript. X.L. performed experiments and provided advice in experimental plans. W.Z. provided advice in experimental plans. T.X. directed all the experiments and participated in experimental design, data analyses, interpretation, and manuscript writing.

Declaration of interests

The authors declare no competing interests.

Supporting Information

Supplementary file 1. PCR primer information in this study.

Figure 6-figure supplement 1. The regulation of m⁶A writers (METTL3&14) to the MDA5-exon1. (A) Relative luciferase activity of pmirGLO-MDA5-exon1 firefly luciferase reporters in MPC cells transfected with si-Ctrl, si-METTL3 or si-METTL14. (B) Schematic diagram of m⁶A site mutation on exon1 of MDA5. 'A' in the predicted methylation motif was replaced by 'U'. (C) Relative luciferase activity of wild-type or mutant pmirGLO-MDA5-exon1 firefly luciferase reporter in MPC cells transfected with si-Ctrl or si-METTL3&14. (D) METTL3 and METTL14 could downregulate GFP expression of mVenus-MDA5-exon1. EPC cells were co-transfected with the mVenus-MDA5-exon1 and vector, METTL3 or METTL14 plasmids. At 48 h post-transfection, the fluorescence intensity and the GFP expression were evaluated by enzyme-labeled instrument and western blotting, respectively. All data presented as the means $\pm$ SE from at least three independent triplicated experiments. **, $p < 0.01$ versus the controls.

Figure 7-figure supplement 1. The regulation of YTHDF readers (YTHDF1&2&3) to the MDA5-exon1. (A) Relative luciferase activity of pmirGLO-MDA5-exon1 firefly luciferase reporters in MPC cells transfected with si-Ctrl, si-YTHDF1, si-YTHDF2 or si-YTHDF3. (B) Relative luciferase activity of wild-type or mutant pmirGLO-MDA5-exon1 plasmids in MPC cells transfected with si-Ctrl or si-YTHDF2&3. (C) YTHDF2 and YTHDF3 could downregulate GFP expression of mVenus-MDA5-exon1. EPC cells were co-transfected with the mVenus-MDA5-exon1 and vector, YTHDF1, YTHDF2 or YTHDF3 plasmids. At 48 h post-transfection, the fluorescence intensity and the GFP expression were evaluated by enzyme-labeled instrument and western blotting, respectively. All data presented as the means $\pm$ SE from at least three independent triplicated experiments. **, $p < 0.01$ versus the controls.

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Editors

Reviewing Editor

John Schoggins

The University of Texas Southwestern Medical Center, Dallas, United States of America

Senior Editor

John Schoggins

The University of Texas Southwestern Medical Center, Dallas, United States of America

Reviewer #1 (Public Review):

This study offers valuable insights into host-virus interactions, emphasizing the adaptability of the immune system. Readers should recognize the significance of MDA5 in potentially replacing RIG-I and the adversarial strategy employed by 5'ppp-RNA SCR V in degrading MDA5 mediated by m6A modification in different species, further indicating that m6A is a conservational process in the antiviral immune response.

However, caution is warranted in extrapolating these findings universally, given the dynamic nature of host-virus dynamics. The study provides a snapshot into the complexity of these interactions, but further research is needed to validate and extend these insights, considering potential variations across viral species and environmental contexts. Additionally, it is noted that the main claims put forth in the manuscript are only partially supported by the data presented.

<https://doi.org/10.7554/eLife.94898.2.sa2>

Reviewer #2 (Public Review):

This manuscript by Geng et al. aims to demonstrate that MDA5 compensates for the loss of RIG-I in certain species, such as teleofish miiuy croaker. The authors use siniperca cheats rhabdovirus (SCR V) and poly(I:C) to demonstrate that these RNA ligands induce an IFN response in an MDA5-dependent manner in m.miiuy derived cells. Furthermore, they show that MDA5 requires its RD domain to directly bind to SCR V RNA and to induce an IFN response. They use in vitro synthesized RNA with a 5'triphosphate (or lacking a 5'triphosphate as a control) to demonstrate that MDA5 can directly bind to 5'-

triphosphorylated RNA. The second part of the paper is devoted to m6A modification of MDA5 transcripts by SCR V as an immune evasion strategy. The authors demonstrate that the modification of MDA5 with m6A is increased upon infection and that this causes increased decay of MDA5 and consequently a decreased IFN response.

- One critical caveat in this study is that it does not address whether ppp-SCR V RNA induces IRF3-dimerization and type I IFN induction in an MDA5 dependent manner. The data demonstrate that mmiMDA5 can bind to triphosphorylated RNA (Fig. 4D). In addition, triphosphorylated RNA can dimerize IRF3 (4C). However, a key experiment that ties these two observations together is missing.
- Specifically, although Fig. 4C demonstrates that 5'ppp-SCR V RNA induces dimerization (unlike its dephosphorylated or capped derivatives), this does not prove that this happens in an MDA5-dependent manner. This experiment should have been done in WT and siMDA5 MKC cells side-by-side to demonstrate that the IRF3 dimerization that is observed here is mediated by MDA5 and not by another (unknown) protein. The same holds true for Fig. 4J.
- Fig 1C-D: these experiments are not sufficiently convincing, i.e. the difference in IRF3 dimerization between VSV-RNA and VSV-RNA+CIAP transfection is minimal.
- Fig. 2N and 2O: why did the authors decide to use overexpression of MDA5 to assess the impact of STING on MDA5-mediated IFN induction? This should have been done in cells transfected with SCR V or polyIC (as in 2D-G) or in infected cells (as in 2H-K). In addition, it is a pity that the authors did not include an siMAVS condition alongside siSTING, to investigate the relative contribution of MAVS versus STING to the MDA5-mediated IFN response. Panel O suggests that the IFN response is completely dependent on STING, which is hard to envision.
- Fig. 3F and 3G: where are the mock-transfected/infected conditions? Given that ectopic expression of hMDA5 is known to cause autoactivation of the IFN pathway, the baseline ISG levels should be shown (ie. In absence of a stimulus or infection). Normalization of the data does not reveal whether this is the case and is therefore misleading.
- Fig. 4F and 4G: can the authors please indicate in the figure which area of the gel is relevant here? The band that runs halfway the gel? If so, the effects described in the text are not supported by the data (i.e. the 5'OH-SCR V and 5'pppGG-SCR V appear to compete with Bio-5'ppp-SCR V as well as 5'ppp-SCR V).
- My concerns about Fig. 5 remain unaltered. The fact that MDA5 is an ISG explains its increased expression and increased methylation pattern. The authors should at the very least mention in their text that MDA5 is an ISG and that their observations may be partially explained by this fact.

<https://doi.org/10.7554/eLife.94898.2.sa1>

Reviewer #3 (Public Review):

In this manuscript, the authors explored the interaction between the pattern recognition receptor MDA5 and 5'ppp-RNA in the Miiuy croaker. They found that MDA5 can serve as a substitute for RIG-I in detecting 5'ppp-RNA of Siniperca cheilinus rhabdovirus (SCR V) when RIG-I is absent in Miiuy croaker. Furthermore, they observed MDA5's recognition of 5'ppp-RNA in chickens (*Gallus gallus*), a species lacking RIG-I. Additionally, the authors documented that MDA5's functionality can be compromised by m6A-mediated methylation and degradation of MDA5 mRNA, orchestrated by the METTL3/14-YTHDF2/3 regulatory network in Miiuy croaker during SCR V infection. This impairment compromises the innate antiviral immunity of fish, facilitating SCR V's immune evasion. These findings offer valuable insights into the adaptation and functional diversity of innate antiviral mechanisms in vertebrates.

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Author response:

The following is the authors' response to the original reviews.

eLife assessment

The authors present evidence suggesting that MDA5 can substitute as a sensor for triphosphate RNA in a species that naturally lacks RIG-I. The key findings are potentially important for our understanding of the evolution of innate immune responses, but the evidence is incomplete, as additional biochemical and functional experiments are needed to unambiguously assign MDA5 as a bona fide sensor of triphosphate RNA in this model. This also leaves the title as overstating its case.

We would like to thank the editorial team for these positive comments on our manuscript and the constructive suggestions to improve our manuscript. According to the suggestions and valuable comments of the referees, we have added substantial amounts of new data and analysis to substantiate our claims, and the manuscript, including the title, has been carefully revised to better reflect our conclusions. We are now happy to send you our revised manuscript, we hope the modified manuscript addresses your and the reviewers' concerns satisfactorily and is suitable for publication in eLife now.

Reviewer #1 (Public Review):

This study offers valuable insights into host-virus interactions, emphasizing the adaptability of the immune system. Readers should recognize the significance of MDA5 in potentially replacing RIG-I and the adversarial strategy employed by 5'ppp-RNA SCR in degrading MDA5 mediated by m6A modification in different species, further indicating that m6A is a conserved process in the antiviral immune response.

However, caution is warranted in extrapolating these findings universally, given the dynamic nature of host-virus dynamics. The study provides a snapshot into the complexity of these interactions, but further research is needed to validate and extend these insights, considering potential variations across viral species and environmental contexts.

We concur with the viewpoint that virus-host coevolution complicates the derivation of universal conclusions. To address this challenge, we incorporated additional experiments and data based on the suggestions of the reviewers. These experiments were carried out across diverse models, including two distinct vertebrate species (*M. miiuy* and *G. gallus*), two different viruses (SCR and VSV), and the synthesis of corresponding 5'ppp-RNA probes. We believe that these supplementary data bolster the evidence supporting the immune replacement role of MDA5 in the recognition of 5'ppp-RNA in RIG-I deficient species (Figure 1C-1E, Figure 2O and 2P, Figure 4). Moreover, we have duly incorporated references in both the introduction and discussion sections to further support our conclusion that MDA5 in *T. belangeri*, a mammal lacking RIG-I, possesses the ability to detect RNA viruses posed as RIG-I agonists (doi: 10.1073/pnas.1604939113). Lastly, meticulous revisions have been undertaken in the manuscript, including adjustments to the title, to ensure harmonization with our research outcomes.

Reviewer#2 (Public Review):

This manuscript by Geng et al. aims to demonstrate that MDA5 compensates for the loss of RIG-I in certain species, such as teleost fish miiuy croaker. The authors use siniperca cheati rhabdovirus (SCR) and poly(I:C) to demonstrate that these RNA ligands induce an IFN response in an MDA5-dependent manner in M. miiuy derived cells. Furthermore, they

show that MDA5 requires its RD domain to directly bind to SCR V RNA and to induce an IFN response. They use in vitro synthesized RNA with a 5'triphosphate (or lacking a 5'triphosphate as a control) to demonstrate that MDA5 can directly bind to 5'-triphosphorylated RNA. The second part of the paper is devoted to m6A modification of MDA5 transcripts by SCR V as an immune evasion strategy. The authors demonstrate that the modification of MDA5 with m6A is increased upon infection and that this causes increased decay of MDA5 and consequently a decreased IFN response.

The key message of this paper, i.e. MDA5 can sense 5'-triphosphorylated RNA and thereby compensate for the loss of RIG-I, is novel and interesting, yet there is insufficient evidence provided to prove this hypothesis. Most importantly, it is crucial to test the capacity of in vitro synthesized 5'-triphosphorylated RNA to induce an IFN response in MDA5-sufficient and -deficient cells. In addition, a number of important controls are missing, as detailed below.

To further support the notion that MDA5 is capable of detecting 5'ppp-RNA in species lacking RIG-I, we conducted additional experiments. Initially, we isolated the RNA from SCR V and VSV viruses. Subsequently, we synthesized 5'ppp-RNA probes that corresponded to the genome termini of SCR V and VSV in vitro. Then, these RNAs were treated with Calf intestinal phosphatase (CIAP) to generate dephosphorylated derivatives. Next, we separately tested the activation ability of various RNAs on IRF3 dimer and IFN response in MKC (*M. miiuy* kidney cell line) and DF-1 (*G. gallus* fibroblast cell line) cells, and determined that the immune activation ability of SCR V/VSV viruses depends on their triphosphate structure (Figure 1C-1E, Figure 4C and 4J). In addition, the knockdown of MDA5 inhibited the immune response mediated by SCR V RNA (Figure 2P and 2Q). Finally, we incorporated essential experimental controls (Figure 4B and 4I). We think that the inclusion of these supplementary experimental data significantly enhances the credibility and further substantiates our hypothesis.

The authors describe an interaction between MDA5 and STING which, if true, is very interesting. However, the functional implications of this interaction are not further investigated in the manuscript. Is STING required to relay signaling downstream of MDA5?

To better explore the role of STING in MDA5 signal transduction, we constructed a STING expression plasmid and synthesized specific siRNA targeting STING. Next, we found that co-expression of STING and MDA5 significantly enhance MDA5-mediated IFN-1 response during SCR V virus infection (Figure 2N). Conversely, silencing of STING expression restored the MDA5-mediated IFN-1 response (Figure 2O). These findings provide important evidence for the critical involvement of STING in the immune signaling cascade mediated by MDA5 in response to 5'ppp-RNA viruses.

The second part of the paper is quite distinct from the first part. The fact that MDA5 is an interferon-stimulated gene is not mentioned and complicates the analyses (i.e. is there truly more m6A modification of MDA5 on a per molecule basis, or is there simply more total MDA5 and therefore more total m6A modification of MDA5).

For the experimental data analysis in Figure 5E and 5F, we first compared the m6A-IP group to the input group, and then normalized the control group (IgG group of 5E and Mock group of 5F) to a value of "1". Given the observed variability in MDA5 expression levels within the input group of Mock and SCR V virus-infected cells, our analysis represents the actual m6A content of each MDA5 molecule. To enhance clarity, we have updated the label on the Y-axis in Figure 5E and 5F.

Finally, it should be pointed out that several figures require additional labels, markings, or information in the figure itself or in the accompanying legend to increase the overall clarity of the manuscript. There are frequently details missing from figures that make them difficult to interpret and not self-explanatory. These details are sometimes not even found in the legend, only in the materials and methods section. The manuscript also requires extensive language editing by the editorial team or the authors.

We acknowledge the valuable feedback from the reviewer and have made significant improvements to our manuscript based on the recommendations provided in the "Recommendation for the authors" section. Furthermore, we have conducted a thorough review of the entire article, resulting in substantial enhancements to the format, clarity, and overall readability of our manuscript.

Reviewer#3 (Public Review):

Summary: In this manuscript, the authors investigated the interaction between the pattern recognition receptor MDA5 and 5'ppp-RNA in a teleost fish called Miiuy croaker. They claimed that MDA5 can replace RIG-I in sensing 5'ppp-RNA of Siniperca cheilichthys rhabdovirus (SCRV) in the absence of RIG-I in Miiuy croaker. The recognition of MDA5 to 5'ppp-RNA was also observed in the chicken (Gallus gallus), a bird species that lacks RIG-I. Additionally, they reported that the function of MDA5 can be impaired through m6A-mediated methylation and degradation of MDA5 mRNA by the METTL3/14-YTHDF2/3 regulatory network in Miiuy croaker under SCR infection. This impairment weakens the innate antiviral immunity of fish and promotes the immune evasion of SCR.

Strengths:

These findings provide insights into the adaptation and functional diversity of innate antiviral activity in vertebrates.

Weaknesses:

However, there are some major and minor concerns that need to be further addressed. Addressing these concerns will help the authors improve the quality of their manuscript. One significant issue with the manuscript is that the authors claim to be investigating the role of MDA5 as a substitute for RIG-I in recognizing 5'ppp-RNA, but their study extends beyond this specific scenario. Based on my understanding, it appears that sections 2.2, 2.3, 2.5, 2.6, and 2.7 do not strictly adhere to this particular scenario. Instead, these sections tend to investigate the functional involvement of Miiuy croaker MDA5 in the innate immune response to viral infection. Furthermore, the majority of the data is focused on Miiuy croaker MDA5, with only a limited and insufficient study on chicken MDA5. Consequently, the authors cannot make broad claims that their research represents events in all RIG-I deficient species, considering the limited scope of the species studied.

We agree with the reviewer's perspective that functional analysis of MDA5 in *M. miiuy* may not adequately represent all species lacking RIG-I. To address this concern, we have incorporated additional experimental data utilizing different model systems, including two different vertebrate species (*M. miiuy* and *G. gallus*), two distinct viruses (SCRV and VSV), and the synthesis of two corresponding 5'ppp-RNA probes. While the functional characterization of *G. gallus* MDA5 remains relatively limited compared to *M. miiuy*, our current experimental findings provide support for two key observations. Firstly, the triphosphate structure of the VSV virus is pivotal in activating the innate immune response in *G. gallus* against the virus (Figure 1D and 4J). Secondly, *G. gallus* MDA5 can recognize 5'ppp-RNA (Figure 4I, 4K and 4L). Consequently, although we cannot definitively establish the immune surrogate function of MDA5 in all RIG-I-deficient species, our research data further substantiates this hypothesis.

Moreover, we have adopted a more cautious attitude in summarizing our experimental conclusions, thereby enhancing the rigor of our manuscript language.

The current title of the article does not align well with its actual content. It is recommended that the focus of the research be redirected to the recognition function and molecular mechanism of MDA5 in the absence of RIG-I concerning 5'ppp-RNA. This can be achieved through bolstering experimental analysis in the fields of biochemistry and molecular biology, as well as enhancing theoretical research on the molecular evolution of MDA5. It is advisable to decrease or eliminate content related to m6A modification.

Following the reviewer's recommendations, we have revised the title to emphasize that our main research focus is a teleost fish devoid of RIG-I. Furthermore, we have conducted additional molecular experiments to further elucidate the 5'ppp-RNA recognition function of MDA5 in RIG-I-deficient species. In an attempt to analyze the potential molecular evolution of MDA5 resulting from RIG-I deficiency, we collected MDA5 coding sequences from diverse vertebrates. However, due to multiple independent loss events of RIG-I in fish, fish with or without RIG-I genes in the phylogenetic tree cannot be effectively clustered separately, making it extremely difficult to perform this aspect of analysis. Consequently, we have regrettably opted to forgo the molecular evolution analysis of MDA5.

Our article topic is to reveal an antagonistic phenomenon between fish receptor and RNA viruses. The MDA5 of RIG-I-lost fish has evolved the ability to recognize 5'ppp-RNA virus and mediate IFN response to resist SCRV infection. Conversely, the m6A methylation mechanism endows the SCRV virus with a means to weaken the immune capacity of MDA5. Therefore, we believe that the latter part is an important part of the arms race between the virus and its host, and should be retained.

Additionally, the main body of the writing contains several aspects that lack rigor and tend to exaggerate, necessitating significant improvement.

We appreciate the reviewer's comment and have improved the manuscript addressing the points raised in the "Recommendation for the authors". We have added corresponding experiments to strengthen the verification of the conclusions, and in addition, we are more cautious in summarizing the language of the conclusions.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) The evidential foundation within the Result 1 section appears somewhat tenuous.

Firstly, the author derives conclusions regarding the phenomenon of RIG-I loss in lower vertebrates by referencing external literature and conducting bioinformatics analyses. It is pertinent to inquire whether the author considered fortifying these findings through additional WB/PCR experiments, particularly for evaluating RIG-I expression levels across diverse vertebrates, encompassing both lower and higher orders.

Firstly, the species we analyzed are mostly model species with excellent genomic sequence information in the database. Secondly, the RIG-I protein sequences (at least some domain sequences) are relatively conserved in vertebrates. Therefore, the credibility of evaluating the existence of RIG-I in these species through homology comparison is high. Therefore, we do not intend to conduct additional PCR/WB experiments to confirm this.

Additionally, following the identification of RIG-I loss, the author postulates MDA5 as a substitute of RIG-I, grounding this speculation in the analysis of MDA5 and LGP2 protein structures. It is imperative to address whether the author could enhance the manuscript by supplying expression data for MDA5 and LGP2 across different vertebrates and elucidating further why MDA5 is posited as the compensatory mechanism for RIG-I loss.

Like MDA5, LGP2 is also an interferon-stimulating gene, so they both likely exhibit high sensitivity to viral infections. Therefore, we think that comparing the expression data of these two genes is difficult to evaluate their function. In mammals, the regulatory mechanisms of LGP2 to RIG-I and MDA5 were complicated and ambiguous. To evaluate the potential function of LGP2 in *M. miiuy*, we further constructed LGP2 plasmid and synthesized siRNA targeting LGP2. Then, our results indicate that mmiLGP2 can enhance the antiviral immune response mediated by mmiMDA5 (Figure 1H and 1I), further indicating the regulatory role of mmiLGP2 in RLR signaling, rather than acting as a compensatory receptor for RIG-I.

Also, is it conceivable that other receptors contribute to this compensatory effect in lower vertebrates?

5' triphosphate short blunt-end double-strand RNA is the ligand of RIG-I as contained in the panhandle of negative-strand viral genomes. We mainly focus on the immune recognition and compensatory effects of other receptors on RIG-I loss, and MDA5, as the protein with the most similar structure, first attracted our attention. In addition, IFIT proteins have been reported to recognize triphosphate single-stranded RNA (doi: 10.1038/nature11783). However, we used SCRv and VSV RNA as viral models, both of which have negative stranded genomes and meet the ligand standards of RIG-I, rather than IFIT. Therefore, we excluded the IFIT protein from our research scope.

(2) The article exclusively employs a singular type of 5'PPP-RNA virus and one specific lower vertebrate species, thereby potentially compromising the robustness of the assertion that this phenomenon is prevalent in lower vertebrates. To bolster this claim, could the author consider incorporating data from an alternative 5'PPP-RNA virus and a different lower vertebrate species?

To address this concern, we have incorporated additional experimental data utilizing different model systems, including two different vertebrate species (*M. miiuy* and *G. gallus*) and two distinct viruses (SCRv and VSV). While the functional characterization of *G. gallus* MDA5 remains relatively limited compared to *M. miiuy*, our current experimental findings provide support for two key observations. Firstly, the triphosphate structure of the VSV virus is pivotal in activating the innate immune response in *G. gallus* against the virus (Figure 1D and 4J). Secondly, *G. gallus* MDA5 can recognize 5'ppp-RNA (Figure 4I, 4K and 4L). Consequently, these experimental results further confirmed the conservatism of this immune compensation mechanism.

(3) A nuanced consideration of the statement in Result 5 is warranted. Examination of the results under SCRv infection conditions suggests dynamic fluctuations in MDA5 expression levels, challenging the veracity of the statement implying "increased expression", which contradicts the proposed working model of this article.

Because MDA5 acts as a receptor and plays a recognition immune role in the early stages of virus infection, the expression of MDA5 in the early stage of SCRv infection rapidly increases. In the later stage of infection, the expression of MDA5 may gradually decrease again due to the negative feedback mechanism in the host body to prevent excessive inflammation. However, compared to the uninfected group, the expression of MDA5 was significantly

increased in the SCRV-infected group, so we believe that the term "increased expression" is not a problem. In addition, the m6A mechanism can weaken the function of MDA5, but it still cannot prevent the overall increase of MDA5 expression, which is not contradictory to the working model in this article.

Additionally, the alterations in m6A levels in miiuy croaker under SCRV infection conditions warrant clarification. Could the author employ m6A dot blotting to supplement the findings related to total m6A levels?

Our previous studies (doi: 10.4049/jimmunol.2200618) have suggested that the total m6A level is increased after SCRV infection in miiuy croaker. We cited this conclusion in the discussion of our manuscript.

(4) It would be beneficial if the editors could assist the author in enhancing the language of the manuscript.

We have carefully checked the full article and modified it with Grammarly tools, and we believe that the grammar, format, and readability of our articles have been greatly improved.

Reviewer #2 (Recommendations For The Authors):

Figure 1

(1) Figure 1B - some clarification needs to be added about this figure in the text. It is unclear what the main point is that the authors would like to convey.

What we want to emphasize is that some species with RIG-I, such as zebrafish, have also experienced RIG-I loss events, but have undergone whole genome replication events before the loss, thus preserving a copy of RIG-I. This indicates that loss events of RIG-I are very common in vertebrates and do not occur randomly. We have elaborated on this point in the results and discussion.

(2) Figure 1C - is not very informative other than showing Mm MDA5 and LGP2 side-by-side. It would be more useful to show a comparison of human RIG-I/MDA5 alongside Mm and Gg MDA5. Are there any conserved/shared key residues between hRIG-I/hMDA5 versus mmMDA5?

Homologous proteins are often known to adopt the same or similar structure and function. We have added human RIG-I domain information to this figure (Figure 1F). By comparing the domain information of human RIG-I with *M. miiuy* MDA5 and LGP2, *M. miiuy* MDA5 has a similar structure to human RIG-I, making it most likely to compensate for the missing RIG-I. While *M. miiuy* LGP2 lacks the CARD domain, which is crucial for signal transduction, so we will shift our focus to *M. miiuy* MDA5. In addition, we collected protein sequences of MDA5 and RIG-I from various vertebrates to identify key residues evolved in recognizing 5'ppp-RNA by *M. miiuy* MDA5. However, unfortunately, no potential residues were found during the comparison process.

Figure 2

(1) Figure 2B - It would be important to demonstrate MDA5-Flag expression by immunoblot and compare MDA5-Flag overexpression to endogenous MDA5 expression using the anti-MDA5 antibody from panel 2A. If IF is used, more cells need to be visible in the field.

After transfecting the MDA5 plasmid into MKC, endogenous MDA5 expression was detected using MDA5 antibodies. The results showed a significant increase in MDA5 protein levels, indicating that MDA5 antibodies can specifically recognize MDA5 protein. In addition, we retained the original immunofluorescence images to better demonstrate the subcellular localization of MDA5.

(2) Figure 2C - The 1:1 stoichiometry of MDA5:MAVS (in the absence of any stimulus) is quite surprising. How does the interaction between MDA5 and MAVS change upon stimulation with an RNA ligand (SCRV, poly(I:C))?

We do not believe that the actual stoichiometry between MDA5 and MAVS is what you described as 1:1. In fact, the proportion of proteins in the complex depends on many factors in the experimental results with Co-IP. Firstly, the MDA5 plasmid in this study has a 3 × Flag tag, while the MAVS only has a 1x Myc tag, which makes the antibody more sensitive for detecting MDA5-Flag. In addition, the Co-IP results are also affected by multiple factors such as the type of antibody and the number of recoveries, making it difficult to estimate the actual ratio of MDA5 to MAVS. Based on the above reasons and the fact that the detection of the interaction strength between MDA5 and MAVS after infection seems to be off-topic, we did not continue to explore this point.

(3) Figure 2D - The interaction between MDA5 and STING is a very interesting finding but is not elaborated on in the paper (even though the interaction between MDA5 and STING is mentioned in the abstract). The manuscript would be strengthened if the interaction between MDA5 and STING is further investigated. For example, does the IFN response that is reported in panels 2E to 2H require the presence of STING? Does mmMDA5 signal via STING in response to a DNA ligand?

We appreciate the referee's suggestion to study the mutual influence between MDA5 and STING. We found that co-expression of STING and MDA5 can enhance MDA5-mediated IFN-1 response during SCRV virus infection, while knocking down STING can restore MDA5-mediated IFN-1 (Figure 2N and 2O). This indicates that STING plays an important signaling role in the immune response of MDA5 to RNA viruses. We understand the importance of cGAS/STING pathways in identifying exogenous DNA, so exploring the MDA5 pathway for DNA ligand recognition is an interesting and meaningful perspective. But this seems to be detached from the theme of our article, so we didn't continue to explore this point.

(4) Figures 2F and 2H - the authors demonstrate that SCRV induces a type I IFN response in an MDA5-dependent manner. While SCRV is a single-stranded negative-sense RNA virus that contains 5'ppp-RNA, it cannot be excluded that MDA5 is activated here in response to a double-stranded RNA intermediate of viral origin or even a host-derived RNA whose expression or modification is altered during infection. To demonstrate in an unambiguous manner that MDA5 senses 5'ppp-RNA, it is crucial to use the in vitro synthesized 5'ppp-RNA (and its dephosphorylated derivative as a control) from Fig. 4 in these experiments.

We transfected 5' ppp SCRV and 5' ppp VSV (and their dephosphorylated derivatives) synthesized in vitro into MKC cells and DF-1 cells, respectively. The results showed that 5'ppp-RNAs significantly promoted the formation of IRF3 dimers, while their dephosphorylated derivatives did not (Figure 4C and 4J). In addition, we extracted virus RNA from the SCRV and VSV viruses and dephosphorylated them with Calf intestinal phosphatase (CIAP). These RNAs were transfected into MKC and DF-1 cells and found that the immune response mediated by virus RNAs was much higher than the dephosphorylated form (Figure 1C-1E). The above results indicate that the immune response activated by SCRV and VSV is indeed dependent on

their triphosphate structure. Finally, the IRF3 dimer and IFN induction activated by SCRV RNA can be inhibited by si-MDA5 (Figure 2P and 2Q), further demonstrating the involvement of MDA5 in the immune response mediated by 5'ppp-RNA ligands.

(5) In mice and humans, MDA5 is known to collaborate with LGP2 to jointly induce an IFN response. Does M.miiuy express LGP2? If so, it would be informative to include a siRNA targeting LGP2 in the experiments in panel F. In mammals, LGP2 potentiates the response via MDA5 while it may inhibit RIG-I activation.

M.miiuy express LGP2. We constructed an LGP2 plasmid and synthesized si-LGP2 to investigate the impact of LGP2 on MDA5-mediated immune processes (Figure 1G-1I). The results showed that LGP2 can enhance the IFN response mediated by MDA5 during SCRV virus infection, similar to that in mammals.

(6) Minor comment - Is the poly(I:C) used in this figure high or low molecular weight poly(I:C)? HMW poly(I:C) preferentially stimulates MDA5, while LMW poly(I:C) preferentially stimulates RIG-I.

We used poly(I:C)-HMW as a positive control for activating MDA5. We have modified the relevant information in Figure 2 and its legend.

Figure 3

(1) Figure 3F/G - The normalization in this Figure is difficult to interpret. It would be better to split Figure 3G into 4 separate graphs and include the mock-infected cells alongside the infected samples (as done in Figure 2).

To better demonstrate the function of the RD domain of MDA5 in *M. miiuy*, we have changed the experimental plan, as shown in figure 3F. We detected the induction of antiviral factors by overexpression of MDA5 and MDA5- Δ RD under poly (I:C)-HMW stimulation. This can indicate that the RD domain of MDA5 has a conserved function in the recognition of poly(I:C)-HMW in *M. miiuy*, and can serve as a positive control for the recognition of SCRV virus by the RD domain.

Figure 4

(1) Figure 4B - A number of important controls are missing. Was the immunoprecipitation of RNA successful? This could be shown by running a fraction of the immunoprecipitated material on an RNA gel and/or by showing that the input RNA was depleted after IP. In addition, a control IP (Streptavidin beads without biotinylated RNA) is missing to ensure that MDA5 does not stick non-specifically to the Streptavidin resin.

We appreciate the referee's suggestions. We rerun this experiment and added a non-biomarker RNA IP control group, and the results showed that MDA5 did not adsorb non-specific onto the beads (Figure 4B). In addition, based on the referee's suggestion, we tested the consumption of RNA before and after immunoprecipitation, and the results showed that biotin-labeled RNA, rather than non-biotin-labeled RNA, could be adsorbed by beads, indicating the success of RNA precipitation. However, we think that this is not necessary for the final presentation of the experimental results, so we did not show this in the figure.

(2) Figure 4B - It is unclear why there is such a large molecular weight difference between endogenous MDA5 and MDA5-Flag (110 kDa versus 130/140 kDa). Why is there less MDA5-Flag retrieved than endogenous MDA5?

After careful analysis, we believe that the significant difference in molecular weight between endogenous MDA5 and MDA5 Flag may be due to three reasons. Firstly, MDA5 flag has a 3× Flag tag. Secondly, as shown in the primer table, we constructed MDA5 between the NotI and XbaI cleavage sites in the pcDNA3.1 vector, which are located at the posterior position in the vector. This means that the Flag tag has a certain distance from the starting codon of MDA5, and these sequences on the vector can also be translated and increase the molecular weight of the exogenous MDA5 protein. Finally, in order to facilitate the amplification of the primers, the F-terminal primers of MDA5 contain a small portion of the 3'UTR sequence (excluding the stop codon). These above reasons may have led to significant differences in molecular weight. In addition, in order to supplement important experimental controls, we have conducted a new RNA pull-down experiment as shown in Figure 4B.

(3) Minor point: Figure 4B - please clarify in the figure whether RNA or protein is immunoprecipitated and via which tags.

We have conducted a new RNA pull-down experiment as shown in Fig 4B, and we have clearly labeled the relevant information in the figure.

(4) Figure 4E - the fraction of MDA5 that binds 5'ppp-RNA seems incredibly minor. And why is this experiment done using 5'OH-RNA as a competitor, rather than simply incubating MDA5 and 5'OH-RNA together and demonstrating that these do not form a complex?

The proportion of MDA5 combined with 5'ppp-RNA is influenced by many conditions, including the concentration and purity of the probe and purified protein. In addition, the dosage ratio between the RNA probe and MDA5 protein in the EMSA experiment can also have a significant impact on the results. Therefore, it is not possible to accurately determine the actual binding force between MDA5 and RNA. In the EMSA experimental program, both cold probes (5'ppp-RNA) and mutated cold probes (5'OH-RNA and 5'pppGG-RNA) are crucial for demonstrating the specific binding between MDA5 and 5'ppp-RNA, as they can exclude false positive errors caused by factors such as the presence of biotin in the purified MDA5 protein itself.

(5) Figure 4B/4C/4F - These experiments would be strengthened by including an MDA5 mutant that cannot bind to RNA. These mutants are well-described in mammals. If these residues are conserved, it is straightforward to generate this mutant.

As shown in Figure 3, the MDA5 of *M. miiuy* has an RD domain that can recognize the SCRV virus. We constructed MDA5-ΔRD mutant plasmids with 6x His-tags and purified them for EMSA experiments (Figure 4E). The experimental results further indicate that MDA5, rather than MDA5-ΔRD, can bind to 5'ppp-SCRV (Figure 4G). This further confirms the crucial role of the RD domain in recognizing the 5'ppp-RNA virus.

(6) Minor point: Figure 4E: please clarify in which lanes MDA5 has been added.

Thank you for the referee's suggestion. We have synthesized new 5'ppp-RNA probes (5'ppp-SCRV and their dephosphate derivatives) and rerun this experiment, and relevant information has been added in the Figure (Figure 4F).

Figure 5

(1) Figure 5C - As MDA5 is an interferon-stimulated gene (as shown in panel G/H/I) the increased MDA5 expression could simply explain the increase in the amount of m6A-MDA5 that is immunoprecipitated after infection. Could this figure be improved by doing

a fold change between input vs m6A-IP OR uninfected vs SCRV-infected conditions? This would reveal whether the modification of MDA5 with m6A is really increased after infection.

As shown in Figure 5F below, our data indicates that the proportion of m6A-modified MDA5 does indeed increase after SCRV infection, rather than solely due to the increased expression of MDA5 itself.

(2) Figure. 5E/F - The y-axis is unclear: relative MDA5 m6A levels. Relative to what? Input? Mock infected?

For experiments in Figure 5E/F, we first compared the m6A-IP group with the input group, and then normalized the control group (IgG group of 5E and Mock group of 5F) to “1”. We have replaced the Y-axis name with a clearer one (Figure 5E and 5F).

(3) General comment - It is not mentioned in the text that MDA5 is an interferon-stimulated gene. This would account for the increase in expression (qPCR) after viral infection or poly(I:C) transfection, hence there is no novelty in this finding. In addition, the authors suggest that MDA5 increases at the protein level (by immunoblot) but the increase on these blots is not convincing (figure 5H/5I).

We understand that the increase in expression of MDA5 as an interferon-stimulated gene after viral infection is a common phenomenon. We present this to further validate the m6A sequencing transcriptome data, and to demonstrate that although m6A modification interferes with MDA5 expression during viral infection, it cannot prevent the increase of mRNA level of MDA5. In addition, we rerun the experiment and the results showed that the expression of MDA5 protein can indeed be specifically activated by the SCRV virus and poly(I:C)-HMW.

Figure 6

(1) Figure 6E - What was the MOI of the virus used in this experiment? It is not mentioned in the figure legend.

MOI=5, we have added this point in the figure legend.

Figure 7

(1) Figure 7J - This graphic is somewhat misleading and should be altered to better reflect the conclusions that are drawn in the manuscript. The graphic suggests that MAVS and STING interact, but this is not demonstrated in the paper. In addition, the paper does not demonstrate whether MAVS or STING (or both) are needed downstream of MDA5 to relay signalling. Finally, please draw an arrow from type I IFNs to increased expression of MDA5 to illustrate that MDA5 is an ISG.

Thank you for the referee's suggestion. We have revised the images to more accurately match the conclusions of the manuscript (Figure 7J). Firstly, we have separated the STING protein from the MAVS protein. Secondly, arrows have been used to indicate that MDA5 is an IFN-stimulated gene. Finally, as we have added relevant experiments to demonstrate the importance of MITA protein in the signaling process of MDA5-activated IFN response. In addition, the function of MAVS binding to MDA5 protein and promoting its signal transduction is very conserved, and there is a good research background even in fish with RIG-I deficiency (10.1016/j.dci.2021.104235). Therefore, in Figure 7J, we still chose to bind MAVS to MDA5 protein and use it as a downstream signal transducer of MDA5.

Discussion

(1) *There is very little discussion about METTL and YTHDF proteins in the discussion despite the fact that the last 2 figures are entirely devoted to these proteins.*

Based on the referee's suggestion, we have added relevant content about METTL and YTHDF proteins in the discussion. In addition, the basic mechanism and function of METTL and YTHDF proteins were briefly described in the introduction.

Reviewer #3 (Recommendations For The Authors):

Please refer to the specific suggestions and recommendations. They include proposals for experimental additions, improved methodologies, and suggestions to resolve writing-related concerns.

Major concerns

(1) *I suggest changing the article title to "Functional Replacement of RIG-I with MDA5 in Fish Miiuy Croaker", or a similar title, to make it more focused and closely aligned with the content of the article.*

Following the reviewer's recommendations, we have revised the title to emphasize our primary research subject is a teleost fish that lacks RIG-I. In addition, we have changed "5'ppp-RNA" to "5'ppp-RNA virus" to emphasize the interaction between the virus and the receptor. We believe that the revised title is more in line with the content of the article.

(2) *Due to the inherent limitations in genome sequencing, assembly, and annotation for the Miiuy croaker, comprehensive annotation of immune-related genes remains incomplete. To address this critical gap, it is recommended that authors establish experimental protocols, such as Fluorescence In Situ Hybridization (FISH), to confirm the absence of RIG-I in the Miiuy croaker. They should simultaneously employ MDA5 probes as a positive control for validation purposes.*

The miiuy croaker has good genomic information at the chromosomal level (doi: 10.1016/j.aaf.2021.06.001). In addition, studies have shown that RIG-I is absent in the orders of Perciformes (doi: 10.1016/j.fsirep.2021.100012), while miiuy croaker belongs to the order Perciformes, so it does indeed lose the RIG-I gene. Therefore, we do not intend to use FISH technology to prove this.

(3) *Similarly, it is recommended that the authors first provide evidence of the presence of 5'ppp at the 5' terminus of the genome RNA of SCRV, as demonstrated in the study by Goubau et al. (doi: 10.1038/nature13590, Supplementary figure 1). This evidence is crucial before drawing conclusions about the compensatory role of MDA5 in recognizing 5'ppp RNA viruses, using SCRV as the viral model.*

As suggested by the referee, we extracted SCRV RNA from SCRV virus particles and assessed the 5'-phosphate-dependence of stimulation by SCRV RNA. Calf intestinal phosphatase (CIAP) treatment substantially reduced the stimulatory activity of SCRV RNA in MKC cells of *M. miiuy* (Figure 1C and 1E). In addition, similar results were obtained by transfecting VSV-RNA isolated from VSV virus into DF-1 cells of *G. gallus* (Figure 1D). The above evidences confirm the presence of triphosphate molecular features between SCRV and VSV viruses, and indicating that birds and fish lacking RIG-I have other receptors that can recognize 5'ppp-RNA.

(4) *The 62-nucleotide (nt) 5'ppp-RNA utilized in this study was obtained from Vesicular Stomatitis Virus (VSV). In order to provide direct evidence, it is necessary to include a 62-nt 5'ppp-RNA that is directly derived from SCR V itself.*

We adopted this suggestion and synthesized a 67-nucleotide 5'ppp-SCR V RNA probe. We found that 5'ppp-SCR V activates dimerization of IRF3 and binds to MDA5 of *M. miiuy* in a 5'-triphosphate-dependent manner (Figure 4A-4F).

(5) *Given that RNAs with uncapped diphosphate (PP) groups at the 5' end also activate RIG-I, similar to RNAs with 5'-PPP moieties, and the 5'-terminal nucleotide must remain unmethylated at its 2'-O position to allow RNA recognition by RIG-I, it is necessary for the authors to conduct additional experiments to supplement and validate these two distinguishing features of RIG-I in RNA recognition. This will provide more reliable evidence for the replacement of RIG-I by MDA5 in RNA recognition.*

Thank you for the reviewer's professional suggestions. We understand that exploring the combination of 5'pp-RNA and 2'-O-methylated RNA with MDA5 can further demonstrate the alternative function of MDA5. But we think that the use of 5'ppp-RNA and their dephosphorylation derivatives can fully demonstrate that the MDA5 of *M. miiuy* and *G. gallus* have evolved to recognize 5'triphosphate structure like human RIG-I. Therefore, we do not intend to conduct any additional experiments

(6) *In section 2.3, the authors assert that Miiuy croaker recognizes SCR V through its RD domain. This claim is supported by their data showing that cells overexpressed with the MDA5 Δ RD mutant lost the ability to inhibit SCR V replication. As a result, the authors draw the conclusion that "these findings provide evidence that MDA5 may recognize 5'-triphosphate-dependent RNA (5'ppp-RNA) through its RD domain." However, to strengthen their argument, the authors should first demonstrate that during SCR V infection, MDA5-mediated antiviral immune response is indeed initiated by recognizing the 5'ppp part of the SCR V RNA, rather than the double-strand part (which can exist in ssRNA virus) of the viral RNA, as this is naturally a ligand for MDA5. Additionally, the authors should treat the isolated SCR V RNA with CIP to remove the phosphate group and examine the binding of MDA5 with SCR V RNA before and after treatment. They should also transfect CIP-treated or untreated SCR V RNA into MDA5 knockdown and wild-type MKC cells to investigate the induction of antiviral signaling and levels of viral replication. Finally, the authors should verify the binding ability of the mutants with isolated SCR V RNA, with or without CIP treatment, to determine which domain of MDA5 is responsible for SCR V 5'ppp-RNA recognition.*

We understand the reviewer's concern that MDA5 may be identified by binding to dsRNA in the SCR V virus. Based on the reviewer's suggestion, we extracted SCR V RNA and obtained its dephosphorylated RNA using Calf intestinal phosphatase (CIAP). Next, we transfected them into MDA5-knockdown and wild-type MKC cells, and detected the dimerization of IRF3 and IFN reaction. The results indicate that SCR V RNA does indeed activate immunity in a triphosphate-dependent manner, and knockdown of MDA5 prevents immune activation of SCR V RNA (Figure 1C and 1E, Figure 2P and 2Q). Finally, we synthesized a 5'ppp-SCR V RNA probe and demonstrated that MDA5 binds to 5'ppp-SCR V through the RD domain (Figure 4E-4G). We believe that these results can better demonstrate that MDA5 recognizes 5'ppp-RNA through its RD domain and addresses the concerns of the reviewers.

(7) *Similarly, merely presenting Co-IP data demonstrating the interaction between Miiuy croaker MDA5 and STING in overexpressed EPC cells does not justify the claim that "in vertebrates lacking RIG-I, MDA5 can utilize STING to facilitate signal transduction in the*

antiviral response". This is because interactions observed through overexpression may not accurately reflect the events occurring during viral infection or their actual antiviral functions. To provide more robust evidence, it is essential to conduct functional experiments after STING knockout (or at least knockdown). Furthermore, it is important to note that Miiuy Croaker alone cannot adequately represent all "vertebrates lacking RIG-I".

We found that co-expression of STING and MDA5 can enhance MDA5-mediated IFN-1 response during SCRV virus infection, while knocking down STING can restore MDA5-mediated IFN-1 response (Figure 2N and 2O). This indicates that STING plays an important signaling role in the immune response of MDA5 to RNA viruses. In addition, loss of RIG-I is a common phenomenon in vertebrates, and STING of birds such as chickens (doi: 10.4049/jimmunol.1500638) and mammalian tree shrews (doi: 10.1073/pnas.1604939113) can also bind to MDA5, indicating that STING can indeed play a crucial role in MDA5 signaling in species with RIG-I deficiency. We have added this section to our discussion and elaborated on our observations in more cautious language.

(8) In the manuscript, a series of experiments were conducted using an antibody (Beyotime Cat# AF7164) against endogenous MDA5. The corresponding immunogen for this MDA5 antibody is a recombinant fusion protein containing amino acids 1-205 of human IFIH1/MDA5 (NP_071451.2). However, the amino acid sequences of IFIH1/MDA5 differ substantially between humans and Miiuy croaker, which could introduce errors in the results. Therefore, it is essential to employ antibodies specifically designed for targeting Miiuy croaker's own MDA5 in the experiments.

As shown in Figure 2B, endogenous MDA5 antibodies can detect the MDA5 portion that is forcibly overexpressed by plasmids, suggesting that the MDA5 antibody can indeed specifically recognize the MDA5 protein of *M. miiuy*.

(9) It is recommended to investigate the phosphorylation of IRF3 in order to confirm the downstream signaling pathway during viral infection when MDA5 is knocked down or overexpressed.

Due to the lack of available phosphorylation antibodies for fish IRF3, we used IRF3 dimer experiments to detect downstream signaling (Figure 1C and 1D, Figure 2P, Figure 4C and 4J).

(10) The use of poly I:C as a mimic for dsRNA to investigate MDA5's recognition of 5'ppp-RNA in hosts lacking RIG-I, as well as the examination of the regulatory role of MDA5 m6A methylation upon activation by 5'ppp-RNA, may be inappropriate. Poly I:C does not possess 5'ppp, and while it has been identified as a ligand for MDA5 in various studies, MDA5 cannot serve as a substitute for RIG-I in recognizing poly (I:C). Therefore, the authors should utilize 5'ppp-dsRNA as the mimic and include the corresponding 5'ppp-dsRNA control without a 5'triphosphate as the negative control (both available from InvivoGen). This approach will specifically elucidate the mechanisms involved when MDA5 functions similarly to RIG-I in the recognition of 5'ppp-RNA.

In our study, we used poly(I:C)-HMW, a known dsRNA mimetic that can be preferentially recognized by MDA5 rather than RIG-I, as a positive control for activating MDA5. What we want to demonstrate is that, like poly(I:C)-HMW (positive control), SCRV can also promote MDA5-mediated IFN immunity, further indicating the important role of MDA5 in 5'ppp-RNA virus invasion. We have clearly labeled the type of poly(I:C) in the figures and legends to avoid misunderstandings for readers.

(11) In Figure 2, Figure 3, and Figure 6, the appearance of virus plaques is not readily apparent, and it is necessary to replace these images with clearer photographs. It appears that MKC or MPC cells are not appropriate for conducting plaque assays. To accurately assess viral proliferation, the authors should measure key indicators throughout the process, such as the production of positive-strand RNAs (+RNAs), replication intermediates (RF), and transcription of subgenomic RNAs. This approach is preferable to solely measuring the M and G protein genes from the virus genome as positive results can still be observed in contaminated cells.

As pointed out by the reviewer, we also think that the virus plaque images in Figure 2K and Figure 3D are not clear enough, so we have replaced them with new clear images (Figure 2J and Figure 3D). But we think that other images can clearly display the proliferation of the SCR-V virus, so we did not replace them. In addition, the primers we currently use do measure +RNA, so the replication level of the SCR-V virus can be accurately evaluated without being affected by virus contamination. Because the regions where the two pairs of primers are located belong to the SCR-V-M and SCR-V-G protein genes, we label them as SCR-V-M and SCR-V-G to distinguish between the two pairs of genes. To avoid reader misunderstanding, we have modified the Y-axis label in the figures (Figure 2I and 2K, Figure 3E, Figure 6E and 6O).

(12) There is a substantial disparity in the molecular size of M. miiuy MDA5 between endogenous and exogenously expressed proteins, as shown in Figure 2A and 2C-D. Please provide clarification.

Please refer to the response to Reviewer 2's question regarding Figure 4B above.

(13) The manuscript incorporates the evolutionary perspective, but lacks specific evolutionary analysis. Thus, it is essential to include relevant analysis to comprehend the evolutionary dynamics and positive selection on MDA5 and LGP2 in the absence of RIG-I in Miiuy croaker. This can be achieved through theoretical calculations using appropriate algorithms, such as the branch models and branch-site models based on the maximum-likelihood method implemented in the phylogenetic analysis by maximum likelihood (PAML) package.

In fact, we have analyzed the molecular evolution of MDA5 and LGP2. Unfortunately, even when analyzing only the MDA5/LGP2 CDS sequences in fish, we found that the topologies of gene trees of MDA5/LGP2 were largely consistent with the species tree. Thus, species with or without RIG-I in the gene trees cannot effectively separate clusters, making it extremely difficult to analyze the molecular evolution of MDA5/LGP2 caused by RIG-I deficiency. Consequently, we gave up this aspect of analysis.

(14) If the narrative regarding m6A methylation goes beyond the activation of MDA5 through recognition of 5'ppp-RNA and represents a regulatory mechanism for all MDA5 activation events, it is not relevant to the theme of "An arms race under RIG-I loss: 5'ppp-RNA and its alternative recognition receptor MDA5." Therefore, all investigations in this paper should focus solely on events when MDA5 recognizes 5'ppp-RNA. Any data associated with the broader regulatory mechanisms and m6A methylation of MDA5 should be excluded from this manuscript and instead be included in a separate study dedicated to exploring this specific topic.

Our theme aims to showcase RNA viruses, rather than an interaction between 5'ppp-RNA and host virus receptors, which our current topic cannot accurately express. Therefore, we made two main changes: firstly, we limited the study species to M. miiuy, although some studies on the functional substitution of MDA5 for RIG-I involved birds. Secondly, change "5'ppp-RNA" to

“5’ppp-RNA virus”. We believe that the revised title is more in line with our current research contents.

(15) *The running title appears to be hastily done.*

We modified it to “MDA5 recognizes 5’ppp-RNA virus in species lacking RIG-I”.

(16) *There are many descriptions that are not strongly related to the main theme of the article in the introduction section, making it lengthy and fragmented. Please focus on the research background of RIG-I and MDA5, including their structures, functions, and regulatory mechanisms, as well as the research progress on the compensatory effect of MDA5 in the absence of RIG-I and its evolutionary adaptation mechanism in other species.*

Based on the suggestions of the reviewers, we have removed some of the less relevant content in the introduction and added research progress on the compensatory effect of MDA5 in the evolutionary adaptation mechanism of tree shrews in the absence of RIG-I.

(17) *Lines 149-156 in the "Results" section include content that resembles an "Introduction" It is important to avoid duplicating information in the results section. Therefore, the authors are encouraged to revise this paragraph to ensure conciseness in the article.*

We have streamlined this section to enhance the article's conciseness and clarity.

(18) *In the "Results" section, at line 177, the authors assert, "As depicted in Figure 1F-1H," which should be corrected to Figure 2F-2H. Furthermore, the y-axis of the two figures on the right-hand side of Figure 2H represents the ISG15 genes. At line 182, "as demonstrated in Figure 1I-1L," should be revised as "as illustrated in Figure 2I-2L". The authors demonstrated a lack of attention to detail.*

Thank you to the reviewer for pointing out our errors, and we have made the necessary corrections.

(19) *In lines 197-198, the authors stated that "MDA5-ΔRD showed an inability to interact with SCRV." However, Figure 3D did not reveal any significant difference, thus it is advisable to repeat this experiment at least once.*

We have replaced this virus spot image with a new one (Figure 3D).

(20) *In lines 200-201 of the "2.3 RD domain is required for MDA5 to recognize SCRV" section, the authors report that the expression of antiviral genes was induced by the overexpression of both MDA5 and MDA5-ΔRD, even in the absence of infection (Figure 3F). Why does the expression of antiviral genes increase in the absence of viral RNA stimulation? Please provide a reasonable explanation.*

In the absence of viral infection, overexpression of viral receptor proteins may still transmit erroneous signaling, affecting the body's immunity. We speculate that due to the preservation of the CARD domain by MDA5 and MDA5-ΔRD, they can still induce the expression of antiviral factors without ligands, although this induction effect is much smaller than that of viral infection. However, in order to better demonstrate the function of the RD domain of MDA5 in *M. miiuy*, we have changed the experimental plan, as shown in the figure 3F. We detected the induction of antiviral factors by overexpression of MDA5 and MDA5-ΔRD under poly (I:C)-HMW stimulation. This can indicate that the RD domain has a conserved function in the

recognition of poly(I:C)-HMW in *M. miiuy*, and can serve as a positive control for the recognition of SCR virus invasion by the RD domain of MDA5.

(21) Please provide the GeneBank accession number of *M. miiuy* MDA5.

The GeneBank accession number of *M. miiuy* MDA5 was added in the section 4.5 plasmids construction.

(22) The content of lines 228-233 in the "Results" section bears resemblance to that of the "Introduction." To ensure the avoidance of information duplication, it is recommended to remove this paragraph from the results section.

This section has been streamlined.

(23) The bands of *mmiMDA5* in the 5'ppp-RNA and dsRNA lanes in Figure 4B are weak and almost unobservable. Please replace them with clear images.

We have rerun this experiment and replaced the images (Figure 4B).

(24) In Figure 5G and at line 253, there are only results presented for the SCR virus infection group, while no results are shown for the control group. This raises the question of why the control group results are missing. It is necessary to provide a reasonable explanation or correction for this issue.

The "0 h" infection time point of the SCR virus is the control group, and we have replaced it with a more intuitive image (Figure 5G).

(25) In Figure 7C, it would be necessary to include the western blot result of YTHDF protein expression in order to verify the efficiency of YTHDF siRNA.

In fact, we have attempted to detect the endogenous expression of YTHDF protein using available commercial antibodies. Unfortunately, only the YTHDF2 antibody can specifically recognize the endogenous protein expression of YTHDF2 in *M. miiuy*. In addition, the knockdown effect of si-YTHDF2 has been validated by YTHDF2 antibody (doi: 10.4049/jimmunol.2200618).

(26) In line 422 of the "4.3 Cell culture and treatment" section, the paragraph raises a question regarding the nature of *Miiuy* croaker kidney cells (MKCs) and spleen cells (MPCs) - whether they are cell lines or freshly isolated cells (or primary cultures) derived from kidney and spleen tissues. If these cells are indeed cell lines, it is requested to provide detailed information about the sources and properties of the cells (such as whether they are epithelial cells or other mixed cell types) and the generations of propagation. Alternatively, if the cells were freshly isolated or primary cultures obtained from fish, the method for cell isolation should be provided. The source and stability of cells are extremely important for ensuring the repeatability and reliability of experimental outcomes.

M. miiuy kidney cells (MKCs) and spleen cells (MPCs) are cell lines derived from the kidney and spleen tissues of *M. miiuy*, with passages ranging from 20 to 40 times. These details have been incorporated into section 4.3.

(27) There are many inaccurate descriptions in the text, which employ concepts that are too broad. These descriptions need to be narrowed down to specific species or objects. Here are a few examples, along with the necessary revisions. Other similar instances

should also be revised accordingly. For instance, in line 119, "fish MDA5" should be changed to "Miiuy croaker MDA5." Similarly, in line 166, "fish MDA5-mediated signaling pathway" should be changed to "Miiuy croaker MDA5-mediated signaling pathway." In line 174, "fish MDA5" should be revised to "Miiuy croaker MDA5." Additionally, in line 185, "antiviral responses of teleost" should be changed to "antiviral responses of Miiuy croaker." In line 197, "interact with SCRV" should be revised to "interact with 5'ppp-RNA of SCRV." In line 337, "loss of RIG-I in the vertebrate" should be modified to "loss of RIG-I in Miiuy croaker and chicken." Similarly, in line 338, "MDA5 of fish" should be changed to "MDA5 of Miiuy croaker." Lastly, in line 348, "RIG-I deficient vertebrates" should be revised to "RIG-I deficient Miichthys miiuy and Gallus gallus."

Thank you for the reviewer's suggestions. We have made revisions to these inaccurate descriptions and reviewed the entire manuscript to address similar statements with broad concepts.

(28) Finally, it should be noted that a similar discovery has already been reported in tree shrews (Ling Xu, et al., Proc Natl Acad Sci., 2016, 113(39):10950-10955). This article shares similarities with that research report, therefore it is necessary to discuss in detail the relationship between the two in the discussion and compare and analyze the evolutionary patterns of MDA5 from it.

Based on the reviewer's suggestions, we have compared the similarities and differences between these two reports during the discussion and analyzed the evolutionary dynamics of MDA5 in these vertebrates lacking RIG-I.

Minor concerns:

Thank you to the reviewer for their meticulous examination to our manuscript, we have made revisions to the following suggestions.

(1) At line 120, the sentence "SCRV(one 5'ppp-RNA virus)" should have a space between "SCRV" and "(one 5'ppp-RNA virus)". Please make this correction.

Corrected.

(2) At lines 147-148, the sentence "However, the downstream gene of TOPORSa is missing a RIG-I" is not accurate and needs modification.

We have modified this sentence.

(3) At line 184, "findings indicate" should be corrected to "findings indicated".

Corrected.

(4) At line 189, "a 5'ppp-RNA virus" should be deleted and the text seems redundant.

Deleted.

(5) At line 198, "replication. (Figure 3C-3E)", please remove the punctuation between "replication" and "(Figure 3C-3E)".

Corrected.

(6) At line 416 in "Materials and methods" section, "4.2 Sample and challenge" should be corrected to "4.2 Fish and challenge".

Corrected.

(7) At line 419, the authors state that "The experimental procedure for SCRIV infection was performed as described", please briefly describe the SCRIV infection method and the infectious dose.

Based on the reviewer's suggestions, we have added relevant descriptions of SCRIV infection in section 4.2.

(8) There are several formatting issues in the "Materials and Methods" section. For instance, in line 424, there is no space between the number and letter in "100 µg/ml" and "26 °C" should be corrected to "26 °C". Additionally, in line 430, "Cells" should be corrected to "cells".

Corrected.

(9) At line 446, "50 ng/ul" and "100 mU/ul" should be corrected to "50 ng/µl" and "100 mU/µl".

Corrected.

(10) At line 459, "primers 1)" should be corrected to "primers".

Corrected.

(11) At lines 461-464, the description "For protein purification, MDA5 plasmids with 6× His tag was constructed based on pcDNA3" seems to be no direct logical connection between protein purification and the plasmid construction. Please make the necessary corrections.

Corrected.

(12) At line 548, "cytoplasmic" should be corrected to "Cytoplasmic".

Corrected.

(13) At line 549, "5× 107" should be corrected to "5 × 10⁷".

Corrected.

(14) At line 557, "MgCl2" should be corrected to "MgCl₂".

Corrected.

(15) At line 558, "6 %" should be corrected to "6%".

Corrected.

(16) At line 565, "50µg" should be corrected to "50 µg".

Corrected.

(17) At line 571, "300{plus minus}50 bp." should be corrected to "300 {plus minus} 50 bp."

Corrected.

(18) At lines 592-593, the sentence "After several incubations, the m6A level was quantified colorimetrically at a wavelength of 450 nm" does not read smoothly, please improve it.

Revised.

(19) At line 786, "MDA5 recognize" should be corrected to "MDA5 recognized".

Corrected.

(20) At lines 788 and 798, "Pulldown" should be corrected to "Pull-down".

Corrected.

(21) At lines 790 and 796, "bluestaining" should be corrected to "blue staining".

Deleted.

(22) At line 825, "SCRV and infection" should be corrected to "SCRV infection".

Corrected.

(23) At lines 826-827, "SCRV (H) and poly(I:C) (I) infection" should be corrected to "SCRV infection (H) and poly(I:C) stimulation (I)".

Corrected.

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